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Optimal Error Estimates for Chebyshev Approximations of Functions with Endpoint Singularities in Fractional Spaces

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Abstract
In this paper, we introduce some new definitions and more general results of fractional spaces in order to deal with functions with endpoint singularities. Based on this theoretical framework, we derive optimal decay rates for Chebyshev expansion coefficients by applying the uniform upper bounds of generalized Gegenbauer functions of fractional degree (GGF-Fs). This enables us to further present the optimal L^∞-estimates and L^2-estimates of the Chebyshev polynomial approximations. In particular, we provide point-wise error estimates and the precise upper and lower bounds for $u(x) = (1+x)^\alpha$, $\alpha > 0$ on $\bar{\Omega} = [-1, 1]$ in L^∞-norm. Moreover, we also discuss the extension of our main results to optimal error estimates of the related Chebyshev interpolation and quadrature measured in various norms at Chebyshev–Gauss points. Numerical results demonstrate the perfect coincidence with the error estimates. Indeed, the analysis techniques can enrich the theoretical foundation of p and hp methods for singular problems.

Keywords Approximation by Chebyshev polynomials · Optimal estimates · Singular functions with endpoint singularities · Fractional spaces

Mathematics Subject Classification Primary 41A10 · 41A25 · 41A50 · 65N35 · 65M60

1 Introduction

It is well-known that polynomial approximation and orthogonal expansions are essential ingredients of numerical analysis, which are also used as the basic means of approximation in most areas of scientific computing ([7, 11, 19, 37, 40] and references therein). In recent

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years, spectral and spectral-element methods, or the p-version and hp-version of the finite element methods have attracted more and more interest both in theory and computational practice, especially in the field of solving solutions to many practical applications with weakly singular solutions, such as in non-smooth domains, with non-matching boundary conditions, in integral equations with weakly singular kernels, and in fractional differential equations (see, e.g., [15, 16, 25, 45, 49]). As for most fractional differential equations (FDEs) which model many physical processes have singular solutions. Due to the lack of regularities, it is often difficult to use standard numerical methods to solve fractional differential problems with sharp accuracy. In Chen and Shu [17], an extended technique was provided to recover exponential accuracy for functions with endpoint singularities using point values on standard collocation points. They [18] further developed a technique to recover exponential accuracy from the first N Fourier coefficients of functions which are analytic in the open interval but have unbounded derivative singularities at endpoints.

For initial value problems or initial boundary value problems whose solutions display a typical weak singularity at the initial time $t = 0$, Stynes et al. [38] presented a new analysis of a standard finite difference method for problems with initial singularities. This analysis included considerations for uniform meshes, graded meshes in time, as well as new stability and consistency bounds. Then Chen and Stynes [13] generalised Alikhanov's high-order scheme to nonuniform meshes and gave detailed analysis. Recently, Kopteva and Stynes [22] developed the posteriori theory for the extension of time-fractional FDEs containing several fractional derivatives. An adaptive algorithm based on this theory is tested extensively and shown to yield accurate numerical solutions on the meshes generated by the algorithm.

In general, the presence of corner singularities caused by non-smoothness of the boundary destroys the optimal rate of convergence ([24, 31]). Here, we present an example of the following boundary value problem with Ω being a polygonal domain with several corners (Fig. 1) (cf. [6, p. 45]). The Poisson equation we consider is defined as follows:

$$-\Delta u = f \text{ in } \Omega, \quad u = 0 \text{ on } \partial\Omega, \tag{1.1}$$

and the solutions are given in the following form:

$$g(x)|x-\theta|^{\rho}, \quad \rho \geq 0,$$

where $g(x)$ is a sufficiently smooth function about the region point x (possibly the variable of time direction), and $|x-\theta|$ ($r = |x-\theta|$ in polar coordinates) is the distance from the region to the singularity. Specifically, for (1.1) and polygonal domains Ω with interior angles $\omega_j \in (0, 2\pi)$. Let us consider each vertex A_j of the polygon Ω with polar coordinates $\left(r_j, \theta_j\right)$ (such that the lines $\theta_j = 0$ and $\theta_j = \omega_j$ coincide with the edges meeting at A_j). We define singular functions V_{sj} as follows:

$$V_{sj}\left(r_j, \theta_j\right) = \begin{cases} r_j^{s\pi/\omega_j} \sin\left(\frac{s\pi}{\omega_j}\theta_j\right) & s\pi/\omega_j \notin \mathbb{N}, \\ r_j^{s\pi/\omega_j}\left(\ln r_j \sin\left(\frac{s\pi}{\omega_j}\theta_j\right) + \theta_j \cos\left(\frac{s\pi}{\omega_j}\theta_j\right)\right) & s\pi/\omega_j \in \mathbb{N}. \end{cases}$$

Here are the formulas to explain how the singularity in the (r_j, θ_j) depends on the angle (cf. [30, p. 7–8]).

To deal with corner singularities of the form $u(x) = |x-\theta|^{\alpha}, \alpha > -1/2, x, \theta \in (-1, 1)$, ($\alpha$ is non-integer), non-uniformly Jacobi-weighted Sobolev spaces, $H^{m,\beta}(\Omega)$ with integer $m \geq 0$ and $\beta > -1$ (see, e.g., [4, 5, 19–21, 37]), are introduced instead of the standard weighted Sobolev space $H_w^m(\Omega)$ with $w(x)$ a weight on $\Omega = (-1, 1)$, which is applied to

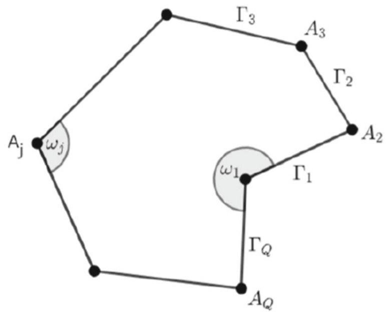

Fig. 1 Polygonal region Ω with boundary vertices A_j, $j = 1, 2, \ldots, Q$ and interior angles ω_j

estimate the orthogonal projection

$$\left\| u - \mathcal{Q}_N^u \right\|_{\mathcal{S}} \le C(N) \|u\|_{H^{m,\beta}(\Omega)},$$

where $\mathcal{Q}_N^u(x)$ is the truncated polynomial of the Gegenbauer or Chebyshev expansion of $u(x)$, $\mathcal{S}$ is a related Besov or Sobolev space. It is worth noting that even for $H^{m,\beta}(\Omega)$ with the weighted norm

$$\|u\|_{H^{m,\beta}(\Omega)} = \left\{ \sum_{l=0}^{m} \int_{-1}^{1} [u^{(l)}(x)]^2 (1-x^2)^{\beta+l} \mathrm{d}x \right\}^{1/2},$$

suboptimal estimates are obtained for $(1+x)^\alpha$-type singular functions with non-integer $\alpha > 0$ (see [12, p. 474] and [27]). In fact, we observe that $u(x) = (1+x)^\alpha \in H^{m,-1/2}(\Omega)$ with $m < 2\alpha + 1/2$, the Chebyshev approximation $\left\| u - \mathcal{Q}_N^u \right\|_{L^2_w(\Omega)}$ with $w(x) = \left(1 - x^2\right)^{-1/2}$ losses an order of the fractional part of $2\alpha + 1/2$, or one order when $2\alpha = k + 1/2$ with nonnegative integer k. For more details, see Liu, Wang, and Li [27].

Recently, Wang [42] considered Clenshaw–Curtis quadrature for integrals with algebraic endpoint singularities. An asymptotic error expansion and convergence rate were derived by combining a delicate analysis of the Chebyshev coefficients of functions with algebraic endpoint singularities and the aliasing formula of Chebyshev polynomials. Liu, Wang, and Li [27] proposed a new theoretical framework built upon fractional Sobolev-type spaces involving generalised Gegenbauer functions of fractional degree (GGF-Fs) for optimal error estimates of Chebyshev polynomial approximations to functions with limited regularity. Subsequently, Liu, Wang, and Wu [28] presented a new fractional Taylor formula for singular functions whose Caputo fractional derivatives are of bounded variation which could be viewed as an "interpolation" between the usual Taylor formulas of two consecutive integer orders and derived the optimal (weighted) L^∞-estimates and L^2-estimates of the Legendre polynomial approximations. Xiang [46] showed optimal convergence rates for Jacobi orthogonal projections with the endpoint singularity $(1 \pm x)^\gamma \ln^\mu(1 \pm x) g(x)$ ($g \in C^\infty[-1, 1]$) and the interior singularity $|x - z_0|^s \ln^\mu |x - z_0| \, g(x)$ ($z_0 \in (-1, 1)$).

More recently, Chen and Shen [14] constructed special classes of orthogonal functions and developed a basic approximation theory for these new orthogonal functions, which they used to solve problems with weakly singular behaviors at the initial time for initial value problems or at one endpoint for boundary value problems. For functions with interior or endpoint singularities, Wang [43] provided a theoretical explanation for the error localization phenomenon of Gegenbauer projections and discussed why the accuracy of Gegenbauer projections is better than that of best approximations except in small neighborhoods of the critical points. Zhang and Boyd [50] investigated the asymptotic coefficients and rigorous error estimates of the Chebyshev expansion of the weak singularity functions on Chebyshev difference basis, Chebyshev polynomials and the quadratic factor basis, which were solved by the least squares method.

In spite of the fact that the optimal decay rate analysis of the Chebyshev expansion coefficients for functions with endpoint singularities has been considered (cf. [27]), there is still a lack of a comprehensive spatial framework and specific theoretical analysis.

In fact, the reference [27] focuses on exact formulas and the decay rate of the Chebyshev expansion coefficients for only a few typical types of functions with endpoint singularities, which provides optimal results. However, the technique used in [27] relies on Taylor expansion for these functions. In this paper, we have enhanced the fractional spatial framework, which includes a wider class of functions. The form of $u(x) = (1 + x)^{\alpha} g(x)$, where $g(x)$ is a sufficiently smooth function, belongs to a subclass within the fractional spatial framework. Unlike the approach presented in [27], our new technique can obtain optimal convergence directly.

We emphasize that the goal of this work is to conduct a more detailed investigation into the optimal error estimates for Chebyshev approximations of functions in fractional spaces with endpoint singularities.

We highlight the main contributions of this work as follows.

(i) We introduce some new definitions and more general results of fractional spaces in order to deal with functions with endpoint singularities. Based on this theoretical framework, we derive the optimal decay rates of the coefficients for functions with endpoint singularities which are expanded in the form of Chebyshev polynomial series and a precise upper bound is given.

(ii) We show the optimal L^{∞}-estimates and L^2-estimates of the Chebyshev polynomial approximations. In particular, we provide point-wise error estimates and the precise upper and lower bounds for $u(x) = (1 + x)^{\alpha}$, $\alpha > 0$ on $\bar{\Omega} = [-1, 1]$ in L^{∞}-norm.

(iii) We present a set of optimal Chebyshev approximation results for the interpolation and quadrature of functions with endpoint singularities at Chebyshev–Gauss points. In fact, these results can enrich the theoretical foundation of p and hp methods for singular problems.

The paper is organised as follows. In Sect. 2, we review the basics of fractional integrals and derivatives, and then we present the uniform bounds of the GGF-Fs. In Sect. 3, we recall the basics of several spaces of functions and give some definitions and more general results of fractional spaces in order to deal with functions with endpoint singularities. We derive the main results on Chebyshev approximation of functions with endpoint singularities in Sect. 4, and discuss in Sect. 5 the extension of the main results to the analysis of interpolation and quadrature. We conclude this paper in Sect. 6.

2 Preliminaries

In this section, we make full preparations for the forthcoming discussions. More precisely, we review the basics of fractional integrals and derivatives, and then recall the generalised Gegenbauer functions of fractional degree(GGF-Fs). Finally, we give the uniform bounds of the GGF-Fs, which will be used in the forthcoming analysis of the decay rate of Chebyshev expansion coefficients.

2.1 Fractional Integrals and Derivatives

Let $\mathbb{N}$ and $\mathbb{R}$ be the set of positive integers and real numbers, respectively. Denote

$$\mathbb{N}_0 := \{0\} \cup \mathbb{N}, \quad \mathbb{R}^+ := \{a \in \mathbb{R} : a > 0\}, \quad \mathbb{R}_0^+ := \{0\} \cup \mathbb{R}^+.$$

We first recall the definition of the Riemann-Liouville (RL) fractional integral (cf. [35, p. 33, p. 44]) as follows:

Definition 2.1 For any $f \in L^1(\Omega)$, the left-sided and right-sided Riemann–Liouville fractional integrals of order $s \in \mathbb{R}^+$ are defined by

$$(I_{a+}^s f)(x) = \frac{1}{\Gamma(s)} \int_a^x \frac{f(y)}{(x-y)^{1-s}} \mathrm{d}y; \quad (I_{b-}^s f)(x) = \frac{1}{\Gamma(s)} \int_x^b \frac{f(y)}{(y-x)^{1-s}} \mathrm{d}y, \quad x \in \Omega,$$

where $\Gamma(\cdot)$ is the usual Gamma function.

The fractional integration has the property (cf. [35, p. 34]):

$$I_{a+}^{s_1} I_{a+}^{s_2} f = I_{a+}^{s_1+s_2} f, \quad I_{b-}^{s_1} I_{b-}^{s_2} f = I_{b-}^{s_1+s_2} f, \quad s_1 > 0, \quad s_2 > 0. \tag{2.1}$$

2.2 Generalised Gegenbauer Functions of Fractional Degree

Much of our discussion later will make use of generalised Gegenbauer functions of fractional degree. Here, the notation and normalization are in accordance with [27, 28]. Recall the hypergeometric function (cf. [2])

$${}_2F_1(a, b; c; z) = \sum_{j=0}^{\infty} \frac{(a)_j (b)_j}{(c)_j} \frac{z^j}{j!}, \quad |z| < 1, \quad a, b, c \in \mathbb{R}, \quad -c \notin \mathbb{N}_0,$$

where for $a \in \mathbb{R}$ and $j \in \mathbb{N}$, the rising factorial in the Pochhammer symbol is defined by

$$(a)_0 = 1; \quad (a)_j = a(a+1)\cdots(a+j-1), \quad \text{for} \quad j \geq 1.$$

Throughout this paper, the generalised Gegenbauer functions (GGF-Fs) we use are defined by (cf. [27, Def. 2.1])

Definition 2.2 For real $\lambda > -1/2$ and real $\nu \geq 0$, the right GGF-F of degree ν is defined by

$${}^{r}G_\nu^{(\lambda)}(x) = {}_2F_1\Big(-\nu, \nu + 2\lambda; \lambda + \frac{1}{2}; \frac{1-x}{2}\Big),$$

for $x \in (-1, 1)$; while the left GGF-F of degree ν is defined by ${}^{l}G_\nu^{(\lambda)}(x) = (-1)^{[\nu]}\, {}^{r}G_\nu^{(\lambda)}(-x)$, where $[\nu]$ is the largest integer $\leq \nu$.

2.3 Uniform Upper Bounds

In order to show the optimal decay rates on the Chebyshev expansion coefficients, we employ the uniform upper bounds of generalised Gegenbauer functions of fractional degree (GGF-Fs) from [27].

Lemma 2.1 **(See** *[27, (4.30)]). For* $\lambda \geq 1$ *and real* $\nu \geq 0$, *we have*

$$\max_{|x| \leq 1} \left\{(1-x^2)^{\lambda-1/2} |{}^{r}G_\nu^{(\lambda)}(x)|\right\} \leq \frac{\Gamma(\lambda+1/2)}{\sqrt{\pi}} \frac{\Gamma((\nu+1)/2)}{\Gamma((\nu+1)/2+\lambda)}.$$

3 Spaces of Functions

In this section, we recall the basics of several spaces of functions that will be used to characterise the regularity of the class of functions of interest, as well as present a useful RL fractional integration by parts formula. Then we give some definitions of the fractional spaces and more general results of spatial relations in order to deal with functions with endpoint singularities.

3.1 Functions of Bounded Variation and Absolutely Continuous Functions

Denote $\Omega = (a, b) \subset \mathbb{R}$ be a finite open interval. Let $C(\bar{\Omega})$ be the classical space of continuous functions, and $\mathrm{AC}(\bar{\Omega})$ the space of absolutely continuous functions on $\bar{\Omega}$. Recall that ([10, Thm. 2.17]): a real function $f(x) \in \mathrm{AC}(\bar{\Omega})$ if and only if f is almost everywhere differentiable in a classical sense, $f'(x)$ belongs to $L^1(\Omega)$, where f' is the weak derivative of f. Also, $f(x)$ has the integral representation([10, (2.29)]):

$$f(x) = f(a) + \int_a^x f'(t)\,\mathrm{d}t, \quad \forall x \in [a, b].$$

Let $\mathrm{BV}(\bar{\Omega})$ be the space of functions of bounded variation on $\bar{\Omega}$. According to [9, Chap. 8]: a function $f(x) \in \mathrm{BV}(\bar{\Omega})$, if there exists a positive constant M such that $V(P, f) := \sum_{i=1}^{k} |f(x_i) - f(x_{i-1})| \leq M$ for each partition $P = \{x_0, x_1, \cdots, x_k\}$ of $[a, b]$. Hence the total variation of f on $\bar{\Omega}$ as $V_{\bar{\Omega}}[f] := \sup\{V(P, f)\}$, where the supreme is taken over all partitions (cf. [3, p. 55] and [23, p. 279]). In general, $V_{\bar{\Omega}}[f]$ could be infinite. According to [33, Thm. 11.23], if $f \in \mathrm{AC}(\bar{\Omega})$, then $V_{\bar{\Omega}}[f] := \int_a^b \left|f'(x)\right| \mathrm{d}x$. Another important class of continuous functions is the space of Lipschitz continuous functions, which are denoted by $H^1(\Omega)$. Every Lipschitz function is absolutely continuous and every absolutely continuous function is of bounded variation, here we show the chain of inclusions $H^1(\Omega) \subseteq \mathrm{AC}(\bar{\Omega}) \subseteq C(\Omega) \subseteq \mathrm{BV}(\bar{\Omega}) \subseteq L^1(\Omega)$. It is evident that $H^1(\Omega) \subseteq \mathrm{AC}(\bar{\Omega})$ but the inverse imbedding is not true. As $f(x) = (x - a)^\beta \in \mathrm{AC}(\bar{\Omega})$, but $(x - a)^\beta \notin H^1(\Omega)$ if $\beta \in (0, 1)$, since the Hölder condition does not hold at the point $x = a$ (cf. [3, p. 74] and [35, p. 3]). Here, we review the definition of the Riemann-Stieltjes (RS) integral (cf. [23, Chap. X] or [3, Chap. 4]): a function f is RS(g)-integrable, if $\int_a^b f \mathrm{d}g < \infty$ for $g \in \mathrm{BV}(\bar{\Omega})$. The space itself also be denoted by RS(g). According to [3, Thm. 4.21], we have the following important property: for all $g \in \mathrm{BV}(\bar{\Omega})$ and f is RS(g)-integrable, then

$$\left| \int_a^b f \mathrm{d}g \right| \leq \|f\|_\infty V_{\bar{\Omega}}[g],$$

where $\|f\|_\infty$ is the L^∞-norm of f on $[a, b]$. Also, if f is continuous and $g \in \mathrm{AC}(\bar{\Omega})$, then $f \in$ RS(g)-integrable, and (cf. [23, Prop.1.6])

$$\int_a^b f(x)\,\mathrm{d}g(x) = \int_a^b f(x) g'(x) \mathrm{d}x. \tag{3.1}$$

3.2 Formula of Integration by Parts

Recall the formula of integration by parts (see, e.g., [23, p. 282]).

Lemma 3.1 *For any $f, g \in \mathrm{BV}(\bar{\Omega})$, we have*

$$\int_a^b f(x)\,\mathrm{d}g(x) = \{f(x)g(x)\}\big|_{a+}^{b-} - \int_a^b g(x)\,\mathrm{d}f(x),$$

where the notation $f(x\pm)$ stands for the right- and left-limit of f at x, respectively.

In particular, if $f, g \in \mathrm{AC}(\bar{\Omega})$, we have

$$\int_a^b f(x)g'(x)\,\mathrm{d}x + \int_a^b f'(x)g(x)\,\mathrm{d}x = \{f(x)g(x)\}\big|_a^b.$$

3.3 Fractional Spaces

To deal with functions with endpoint singularities, we need to give several definitions in order to describe fractional spaces.

Definition 3.1 We say u is of $\mathrm{AC}^k(\bar{\Omega})$ if $u, u', \cdots, u^{(k-1)} \in \mathrm{AC}(\bar{\Omega})$ with integral $k \geq 1$. Also, we denote $\mathrm{AC}^0(\Omega) := L^1(\Omega)$. Equipped with the norm:

$$\|u\|_{\mathrm{AC}^k(\bar{\Omega})} := \sum_{j=0}^{k} \|u^{(j)}\|_{L^1(\Omega)}.$$

Definition 3.2 We say u is of $\mathbb{W}^k(\Omega)$ if $u \in \mathrm{AC}^{k-1}(\bar{\Omega})$ and $u^{(k-1)} \in \mathrm{BV}(\bar{\Omega})$ with integral $k \geq 1$. Also, we denote $\mathbb{W}^0(\Omega) := L^1(\Omega)$. Equipped with the norm:

$$\|u\|_{\mathbb{W}^k(\bar{\Omega})} := \|u\|_{\mathrm{AC}^{k-1}(\bar{\Omega})} + V_{\bar{\Omega}}[u], \quad k \geq 1; \quad \|u\|_{\mathbb{W}^0(\bar{\Omega})} = \|u\|_{L^1(\Omega)}.$$

Remark 3.1 Observing that for $k \in \mathbb{N}_0$, $\mathrm{AC}^{k+1}(\bar{\Omega}) \subseteq \mathbb{W}^{k+1}(\Omega) \subseteq \mathrm{AC}^k(\bar{\Omega}) \subseteq \mathbb{W}^k(\Omega)$. If $k = 0$, we have that $\mathrm{AC}^0(\bar{\Omega}) = \mathbb{W}^0(\Omega) = L^1(\Omega)$. If $k = 1$, we have that $\mathrm{AC}^1(\bar{\Omega}) = \mathrm{AC}(\bar{\Omega})$ and $\mathbb{W}^1(\Omega) = \mathrm{BV}(\bar{\Omega})$. If $k \to \infty$, we have that $\mathrm{AC}^\infty(\bar{\Omega}) = \mathbb{W}^\infty(\Omega) = C^\infty(\Omega)$.

To deal with endpoint singularities, we define the fractional spaces as follows: for $m \in \mathbb{N}_0$ and $s \in (0, 1)$, $\mu := m + s$

$$\mathbb{W}^\mu_{a+}(\Omega) : = \left\{u \in \mathrm{AC}^m(\bar{\Omega}),\ I^{1-s}_{a+} u^{(m)} \in \mathrm{BV}(\bar{\Omega})\right\},$$
$$\mathbb{W}^\mu_{b-}(\Omega) : = \left\{u \in \mathrm{AC}^m(\bar{\Omega}),\ I^{1-s}_{b-} u^{(m)} \in \mathrm{BV}(\bar{\Omega})\right\}.$$

Remark 3.2 Recall that (cf. [35, p. 51]): let $u \in L^1(\Omega)$, then

$$\lim_{\rho \to 0+} I^\rho_{a+} u(x) = u(x), \quad \text{a.e. on } \bar{\Omega}.$$

Similarly, for $u \in L^1(\Omega)$, then

$$\lim_{\rho \to 0+} I^\rho_{b-} u(x) = u(x), \quad \text{a.e. on } \bar{\Omega}.$$

Hence, for $s \to 1-$, we deduce that

$$\mathbb{W}^{m+s}_{a+}(\Omega) = \mathbb{W}^{m+s}_{b-}(\Omega) = \mathbb{W}^{m+1}(\Omega).$$

Thus, we denote

$$\mathbb{W}^{m+1}_{a+}(\Omega) := \lim_{s \to 1-} \mathbb{W}^{m+s}_{a+}(\Omega) = \mathbb{W}^{m+1}(\Omega), \quad \mathbb{W}^{m+1}_{b-}(\Omega) := \lim_{s \to 1-} \mathbb{W}^{m+s}_{b-}(\Omega) = \mathbb{W}^{m+1}(\Omega).$$

Similar to proof of Lemma 2.2 in [26], we have more general results.

Lemma 3.2 *If* $u \in \mathrm{BV}(\bar{\Omega})$ *and* $\rho > 0$, *then* $I_{a+}^{\rho} u(x) \in \mathrm{BV}(\bar{\Omega})$ *and* $I_{b-}^{\rho} u(x) \in \mathrm{BV}(\bar{\Omega})$.

Theorem 3.1 *For* $0 < \mu_1 \leq \mu_2$, *we have*

$$\mathbb{W}_{a+}^{\mu_1}(\Omega) \supseteq \mathbb{W}_{a+}^{\mu_2}(\Omega), \quad \mathbb{W}_{b-}^{\mu_1}(\Omega) \supseteq \mathbb{W}_{b-}^{\mu_2}(\Omega).$$

Proof We denote $\mu_1 = m_1 + s_1$ and $\mu_2 = m_2 + s_2$, respectively. Here $m_1, m_2 \in \mathbb{N}_0$ and $s_1, s_2 \in (0, 1]$.

(i) We consider case: $m_1 = m_2$, thus $0 < s_1 \leq s_2 \leq 1$. If $u \in \mathbb{W}_{a+}^{\mu_2}(\Omega)$, we have

$$u \in \mathrm{AC}^{m_1}(\bar{\Omega}), \quad I_{a+}^{1-s_2} u^{(m_1)} \in \mathrm{BV}(\bar{\Omega}).$$

Using (2.2) and Lemma 3.2, we obtain

$$u \in \mathrm{AC}^{m_1}(\bar{\Omega}), \quad I_{a+}^{1-s_1} u^{(m_1)} = I_{a+}^{s_2-s_1}(I_{a+}^{1-s_2} u^{(m_1)}) \in \mathrm{BV}(\bar{\Omega}).$$

Thus $u \in \mathbb{W}_{a+}^{\mu_1}(\Omega)$, which implies that $\mathbb{W}_{a+}^{\mu_1}(\Omega) \supseteq \mathbb{W}_{a+}^{\mu_2}(\Omega)$. Similarly, we obtain $\mathbb{W}_{b-}^{\mu_1}(\Omega) \supseteq \mathbb{W}_{b-}^{\mu_2}(\Omega)$.

(ii) We consider case: $m_1 < m_2$, thus $m_2 \geq m_1 + 1$. If $u \in \mathbb{W}_{a+}^{\mu_2}(\Omega)$, we have

$$u \in \mathrm{AC}^{m_2}(\bar{\Omega}), \quad I_{a+}^{1-s_2} u^{(m_2)} \in \mathrm{BV}(\bar{\Omega}).$$

Then, using result of case (i) with $\mathrm{AC}^{m_2}(\bar{\Omega}) \subseteq \mathrm{AC}^{m_1+1}(\bar{\Omega}) \subseteq \mathbb{W}^{m_1+1}(\Omega)$, we obtain

$$u \in \mathbb{W}^{m_1+1}(\Omega) = \mathbb{W}_{a+}^{m_1+1}(\Omega) \subseteq \mathbb{W}_{a+}^{m_1+s_1}(\Omega).$$

Therefore $\mathbb{W}_{a+}^{\mu_1}(\Omega) \supseteq \mathbb{W}_{a+}^{\mu_2}(\Omega)$. Similarly, we obtain $\mathbb{W}_{b-}^{\mu_1}(\Omega) \supseteq \mathbb{W}_{b-}^{\mu_2}(\Omega)$. □

Definition 3.3 For some $m, k \in \mathbb{N}_0$, $s \in (0, 1)$ and $\mu := m + s$, assume that $u \in \mathbb{W}_{a+}^{\mu}(\Omega)$ (respectively, $u \in \mathbb{W}_{b-}^{\mu}(\Omega)$), $v \in \mathbb{W}^{k+1}(\Omega)$ with $v(x) = I_{a+}^{1-s} u^{(m)}(x)$ (respectively, $v(x) = I_{b-}^{1-s} u^{(m)}(x)$). Also, we say u is of $\mathbb{W}_{a+}^{\mu,k}(\Omega)$ (respectively, $\mathbb{W}_{b-}^{\mu,k}(\Omega)$). Accordingly, we denote

$$U_{a+}^{\mu,k} := V_{\bar{\Omega}}\left[v^{(k)}\right] + |\sin(s\pi)| \sum_{j=0}^{k} \left|v^{(j)}(a+)\right|,$$

and

$$U_{b-}^{\mu,k} := V_{\bar{\Omega}}\left[v^{(k)}\right] + |\sin(s\pi)| \sum_{j=0}^{k} \left|v^{(j)}(b-)\right|.$$

Remark 3.3 If $k = 0$, $\mathbb{W}_{a+}^{\mu,0}(\Omega) = \mathbb{W}_{a+}^{\mu}(\Omega)$ and $\mathbb{W}_{b-}^{\mu,0}(\Omega) = \mathbb{W}_{b-}^{\mu}(\Omega)$. If $k_1, k_2 \in \mathbb{N}_0$ and $k_1 \leq k_2$, we have $\mathbb{W}_{a+}^{\mu}(\Omega) \supseteq \mathbb{W}_{a+}^{\mu,k_1}(\Omega) \supseteq \mathbb{W}_{a+}^{\mu,k_2}(\Omega)$ and $\mathbb{W}_{b-}^{\mu}(\Omega) \supseteq \mathbb{W}_{b-}^{\mu,k_1}(\Omega) \supseteq \mathbb{W}_{b-}^{\mu,k_2}(\Omega)$.

Remark 3.4 If $k \to \infty$ in Definition 3.3, we have that $v \in C^{\infty}(\Omega)$. Obviously,

$$u(x) = g(x)(x-a)^{\alpha} \in \mathbb{W}_{a+}^{\alpha+1,\infty}(\Omega),$$

where $g(x) \in C^{\infty}(\Omega)$.

Applying the technique of the separation of singularities in [41], we have

$$u(x) = g(x)(x-a)^{\alpha_1}(b-x)^{\alpha_2} = g_1(x)(x-a)^{\alpha_1} + g_2(x)(b-x)^{\alpha_2},$$

where $g(x), g_1(x), g_2(x) \in C^\infty[-1,1]$, $g(a)g(b) \neq 0$ and $\alpha_1, \alpha_2 \geq 0$.

Thus

$$u(x) = u_1(x) + u_2(x), \quad u_1(x) \in \mathbb{W}_{a+}^{\alpha_1+1,\infty}(\Omega), \quad u_2(x) \in \mathbb{W}_{b-}^{\alpha_2+1,\infty}(\Omega).$$

Throughout the paper, we consider the $\Omega = (-1,1)$.

4 Chebyshev Expansions of Functions with Endpoint Singularities

In this section, we derive the main results for Chebyshev approximations of functions with endpoint singularities.

4.1 Exact Formulas and Decay Rate of Chebyshev Expansion Coefficients

The decay rate of the expansion coefficient $|\hat{u}_n^C|$ is essential for the error analysis of Chebyshev expansions in various norms and the related interpolation and quadrature errors, as noted in [29, 47]. Let $\omega(x) = (1-x^2)^{-1/2}$ be the Chebyshev weight function. For any $u \in L^2_\omega(\Omega)$, we expand it in Chebyshev series and denote the partial sum by

$$u(x) = {\sum_{n=0}^{\infty}}' \hat{u}_n^C T_n(x), \quad \pi_N^C u(x) = {\sum_{n=0}^{N}}' \hat{u}_n^C T_n(x), \tag{4.1}$$

where the prime denotes a sum whose first term is halved, and

$$\hat{u}_n^C = \frac{2}{\pi}\int_{-1}^{1} u(x)\frac{T_n(x)}{\sqrt{1-x^2}}\mathrm{d}x = \frac{2}{\pi}\int_0^\pi u(\cos\theta)\cos(n\theta)\mathrm{d}\theta.$$

We now study functions with the endpoint singularities. To fix the idea, we focus on the left endpoint singularity at $x=-1$. For $x=1$, the following theorems hold under corresponding conditions, with $U_{1-}^{\mu,k}$ is in place of $U_{-1+}^{\mu,k}$. Here, we present the main results below.

Theorem 4.1 *For some $m, k \in \mathbb{N}_0$, $s \in (0,1)$ and $\mu := m+s$, assume that $u \in \mathbb{W}^{\mu}_{-1+}(\Omega)$, $v \in \mathbb{W}^{k+1}(\Omega)$ with $v(x) = I_{-1+}^{1-s}u^{(m)}(x)$. Then for $\mu > 1/2$ and $n \geq \mu+k+1$, we have*

$$\begin{aligned}\hat{u}_n^C =& \frac{1}{\sqrt{\pi}\,2^{\mu+k-1}\Gamma(\mu+k+1/2)}\int_{-1}^{1} {}^r\mathcal{G}_n^{(\mu+k)}(x)\,\mathrm{d}v^{(k)}(x) \\ &+ \sum_{j=0}^{k}(-1)^{n+j}\widehat{C}_{n,\mu+j}\sin(\mu\pi)v^{(j)}(-1+),\end{aligned} \tag{4.2}$$

where ${}^r\mathcal{G}_n^{(\mu)}(x) := \big(1-x^2\big)^{\mu-1/2}\, {}^rG_{n-\mu}^{(\mu)}(x)$, and

$$\widehat{C}_{n,\rho} := \frac{2^\rho\Gamma(\rho-1/2)}{\pi^{3/2}}\frac{\Gamma(n-\rho+1)}{\Gamma(n+\rho)}. \tag{4.3}$$

Moreover, we have the following bound:

$$|\hat{u}_n^C| \leq \left\{ \frac{1}{2^{\mu+k-1}\pi} \frac{\Gamma((n-\mu-k+1)/2)}{\Gamma((n+\mu+k+1)/2)} + \frac{2^\mu \Gamma(\mu-1/2)}{\pi^{3/2}} \frac{n\Gamma(n-\mu+1)}{(n-\mu-k)\Gamma(n+\mu)} \right\} U_{-1+}^{\mu,k}. \tag{4.4}$$

Proof As the conditions in Definition 3.3, we derive from [27, (6.13)] with (3.1) that

$$\hat{u}_n^C = \frac{1}{\sqrt{\pi}\, 2^{\mu-1}\Gamma(\mu+1/2)} \left\{ \int_{-1}^{1} {}^r\mathcal{G}_n^{(\mu)}(x) v'(x) \mathrm{d}x + \lim_{x\to -1^+} \left\{ {}^r\mathcal{G}_n^{(\mu)}(x) v(x) \right\} \right\}. \tag{4.5}$$

Now, we prove that

$$\begin{aligned} \hat{u}_n^C =& \frac{1}{\sqrt{\pi}\, 2^{\mu+k-1}\Gamma(\mu+k+1/2)} \int_{-1}^{1} {}^r\mathcal{G}_n^{(\mu+k)}(x) \mathrm{d}v^{(k)}(x) \\ &+ \sum_{j=0}^{k} \frac{1}{\sqrt{\pi}\, 2^{\mu+j-1}\Gamma(\mu+j+1/2)} \left\{ {}^r\mathcal{G}_n^{(\mu+j)}(x) v^{(j)}(x) \right\}\Big|_{x=-1+}. \end{aligned} \tag{4.6}$$

Taking $s = 1$, $\nu = n - \mu - j - 1$ and $\lambda = \mu + j + 1$ in [27, (3.13a)], we obtain for $n \geq \mu + j + 1$,

$${}^r\mathcal{G}_n^{(\mu+j)}(x) = -\frac{1}{2(\mu+j)+1} \left\{ {}^r\mathcal{G}_n^{(\mu+j+1)}(x) \right\}'. \tag{4.7}$$

Since $v \in \mathrm{AC}^k(\bar{\Omega})$, and $v^{(k)} \in \mathrm{BV}(\bar{\Omega})$, using (4.7) and Lemma 3.1, yields

$$\begin{aligned} \int_{-1}^{1} {}^r\mathcal{G}_n^{(\mu)}(x) v'(x) \mathrm{d}x =& -\frac{1}{2\mu+1} \int_{-1}^{1} \left\{ {}^r\mathcal{G}_n^{(\mu+1)}(x) \right\}' v'(x) \mathrm{d}x \\ =& \frac{1}{2\mu+1} \int_{-1}^{1} {}^r\mathcal{G}_n^{(\mu+1)}(x) v''(x) \mathrm{d}x - \frac{1}{2\mu+1} \left\{ {}^r\mathcal{G}_n^{(\mu+1)}(x) v'(x) \right\}\Big|_{x=-1+} \\ =& \cdots = \frac{\Gamma(\mu+1/2)}{2^{k-1}\Gamma(\mu+k-1/2)} \int_{-1}^{1} {}^r\mathcal{G}_n^{(\mu+k-1)}(x) v^{(k)}(x)\, \mathrm{d}x \\ &+ \sum_{j=1}^{k-1} \frac{\Gamma(\mu+1/2)}{2^j \Gamma(\mu+j+1/2)} \left\{ {}^r\mathcal{G}_n^{(\mu+j)}(x) v^{(j)}(x) \right\}\Big|_{x=-1+} \\ =& \frac{\Gamma(\mu+1/2)}{2^k \Gamma(\mu+k+1/2)} \int_{-1}^{1} {}^r\mathcal{G}_n^{(\mu+k)}(x)\, \mathrm{d}v^{(k)}(x) \\ &+ \sum_{j=1}^{k} \frac{\Gamma(\mu+1/2)}{2^j \Gamma(\mu+j+1/2)} \left\{ {}^r\mathcal{G}_n^{(\mu+j)}(x) v^{(j)}(x) \right\}\Big|_{x=-1+}, \end{aligned} \tag{4.8}$$

with the aid of the identity (cf. [32]):

$$\Gamma(\mu+1/2) = \frac{\sqrt{\pi}\,(2\mu-1)!!}{2^\mu}, \quad \mu \in \mathbb{N}_0.$$

Substituting (4.8) into (4.5), then we can get (4.6). From [27, (6.15)], we readily can find

$$\begin{aligned} &\lim_{x\to -1^+} \left\{ {}^r\mathcal{G}_n^{(\mu+j)}(x) \right\} \\ &= (-1)^{n+j} 2^{2(\mu+j)-1} \frac{\sin(\mu\pi)}{\pi} \frac{\Gamma(\mu+j-1/2)\Gamma(\mu+j+1/2)\Gamma(n-\mu-j+1)}{\Gamma(n+\mu+j)}. \end{aligned} \tag{4.9}$$

Thus, using (4.6) and (4.9), we obtain (4.2).

According to (4.3), from straightforward calculations, we find

$$\begin{aligned}\sum_{j=0}^{k}\frac{\widehat{C}_{n,\mu+j}}{\widehat{C}_{n,\mu}}&=\sum_{j=0}^{k}2^j\frac{\Gamma(\mu+j-1/2)}{\Gamma(\mu-1/2)}\frac{\Gamma(n-\mu-j+1)}{\Gamma(n-\mu+1)}\frac{\Gamma(n+\mu)}{\Gamma(n+\mu+j)}\\&=1+\sum_{j=1}^{k}\Bigg(\prod_{i=1}^{j}\frac{2(\mu+i-3/2)}{(n-\mu-i+1)(n+\mu+i-1)}\Bigg)\\&\le 1+\sum_{j=1}^{k}\Big(\frac{\mu+k}{n}\Big)^j<\sum_{j=0}^{k}\Big(\frac{\mu+k}{n}\Big)^j\\&=\frac{n}{n-\mu-k},\end{aligned}\tag{4.10}$$

where we used the fact: for $1\le i\le j\le k$ and $\mu+k<n$,

$$(n-\mu-i+1)(n+\mu+i-1)\ge(n-\mu-k+1)(n+\mu+k-1)\ge 2n.$$

It follows from Lemma (2.1) and (4.2), we can get the bound

$$|\hat{u}_n^C|\le\Bigg\{\frac{1}{2^{\mu+k-1}\pi}\frac{\Gamma((n-\mu-k+1)/2)}{\Gamma((n+\mu+k+1)/2)}+\sum_{j=0}^{k}\widehat{C}_{n,\mu+j}\Bigg\}U_{-1+}^{\mu,k},\tag{4.11}$$

then from (4.10) and (4.11), we obtain (4.4) immediately. □

4.2 L^∞-estimates of Chebyshev Expansions

Here, we estimate the Chebyshev expansion errors in the L^∞-norm for functions with end-point singularities. To fix the idea, we focus on the left endpoint singularity but the results can be extended to the right endpoint setting straightforwardly.

Theorem 4.2 *For some $m,k\in\mathbb{N}_0$, $s\in(0,1)$ and $\mu:=m+s$, assume that $u\in\mathbb{W}^{\mu}_{-1+}(\Omega)$, $v\in\mathbb{W}^{k+1}(\Omega)$ with $v(x)=I_{-1+}^{1-s}u^{(m)}(x)$. Then we have that for $1<\mu\le\mu+k\le N$,*

$$\begin{aligned}\|u-\pi_N^Cu\|_{L^\infty(\Omega)}\le&\Bigg\{\frac{1}{2^{\mu+k-2}(\mu+k-1)\pi}\frac{\Gamma((N-\mu-k)/2+1)}{\Gamma((N+\mu+k)/2)}\\&+\frac{N+1}{N-\mu-k+1}\frac{2^{\mu-1}\Gamma(\mu-1/2)}{\pi^{3/2}(\mu-1)}\frac{\Gamma(N-\mu+2)}{\Gamma(N+\mu)}\Bigg\}U_{-1+}^{\mu,k}.\end{aligned}\tag{4.12}$$

Proof By (4.1) and (4.4), we obtain

$$\begin{aligned}|(u-\pi_N^Cu)(x)|\le\sum_{n=N+1}^{\infty}|\hat{u}_n^C|\le\sum_{n=N+1}^{\infty}&\Bigg\{\frac{1}{2^{\mu+k-1}\pi}\frac{\Gamma((n-\mu-k+1)/2)}{\Gamma((n+\mu+k+1)/2)}\\&+\frac{2^{\mu}\Gamma(\mu-1/2)}{\pi^{3/2}}\frac{n\Gamma(n-\mu+1)}{(n-\mu-k)\Gamma(n+\mu)}\Bigg\}U_{-1+}^{\mu,k}.\end{aligned}\tag{4.13}$$

For brevity, we denote

$$\mathcal{S}_n^{\rho}:=\frac{\Gamma((n-\rho+1)/2)}{\Gamma((n+\rho+1)/2)},\quad\mathcal{T}_n^{\rho}:=\frac{\Gamma((n-\rho+1)/2)}{\Gamma((n+\rho-1)/2)}.$$

A direct calculation leads to the identity:

$$\begin{aligned} \mathcal{T}_n^{\rho} - \mathcal{T}_{n+2}^{\rho} =& \frac{n+\rho-1}{2} \frac{\Gamma((n-\rho+1)/2)}{\Gamma((n+\rho+1)/2)} - \frac{n-\rho+1}{2} \frac{\Gamma((n-\rho+1)/2)}{\Gamma((n+\rho+1)/2)} \\ =& (\rho-1) \frac{\Gamma((n-\rho+1)/2)}{\Gamma((n+\rho+1)/2)} = (\rho-1)\mathcal{S}_n^{\rho}, \end{aligned} \tag{4.14}$$

where we used the identity $z\Gamma(z) = \Gamma(z+1)$. Taking $\rho = \mu + k$ in (4.14), we obtain

$$\begin{aligned} \sum_{n=N+1}^{\infty} \frac{\Gamma((n-\mu-k+1)/2)}{\Gamma((n+\mu+k+1)/2)} &= \sum_{n=N+1}^{\infty} \mathcal{S}_n^{\mu+k} = \frac{1}{\mu+k-1} \sum_{n=N+1}^{\infty} \big\{ \mathcal{T}_n^{\mu+k} - \mathcal{T}_{n+2}^{\mu+k} \big\} \\ &= \frac{1}{\mu+k-1} \big\{ \mathcal{T}_{N+1}^{\mu+k} + \mathcal{T}_{N+2}^{\mu+k} \big\}. \end{aligned} \tag{4.15}$$

We find from [1, (1.1) and Theorem 10] that for $0 \le a \le b$, the ratio

$$R_b^a(z) := \frac{\Gamma(z+a)}{\Gamma(z+b)}, \quad z \ge 0, \tag{4.16}$$

is decreasing with respect to z. As $\mu + k - 1 > 0$, we have

$$\mathcal{T}_{N+2}^{\mu+k} = R^0_{\mu+k-1}(1 + (N-\mu-k+1)/2) \le R^0_{\mu+k-1}(1 + (N-\mu-k)/2) = \mathcal{T}_{N+1}^{\mu+k}. \tag{4.17}$$

Hence, follows from (4.15) and (4.17)

$$\sum_{n=N+1}^{\infty} \frac{\Gamma((n-\mu-k+1)/2)}{\Gamma((n+\mu+k+1)/2)} \le \frac{2}{\mu+k-1} \frac{\Gamma((N-\mu-k)/2+1)}{\Gamma((N+\mu+k)/2)}. \tag{4.18}$$

We now show that

$$\begin{aligned} &\sum_{n=N+1}^{\infty} \frac{\Gamma(n-\mu+1)}{\Gamma(n+\mu)} \\ &= \frac{1}{2(\mu-1)} \sum_{n=N+1}^{\infty} \left\{ (n+\mu-1)\frac{\Gamma(n-\mu+1)}{\Gamma(n+\mu)} - (n-\mu+1)\frac{\Gamma(n-\mu+1)}{\Gamma(n+\mu)} \right\} \\ &= \frac{1}{2(\mu-1)} \sum_{n=N+1}^{\infty} \left\{ \frac{\Gamma(n-\mu+1)}{\Gamma(n+\mu-1)} - \frac{\Gamma(n-\mu+2)}{\Gamma(n+\mu)} \right\} \\ &= \frac{1}{2(\mu-1)} \frac{\Gamma(N-\mu+2)}{\Gamma(N+\mu)}, \end{aligned}$$

together with (4.13) and (4.18) implies (4.12). □

Remark 4.1 Recall that (cf. [32, (5.11.13)]): for $a < b$,

$$\frac{\Gamma(z+a)}{\Gamma(z+b)} = z^{a-b} + \frac{1}{2}(a-b)(a+b-1)z^{a-b-1} + O(z^{a-b-2}), \quad z \gg 1. \tag{4.19}$$

Thus, under the conditions of Theorem 4.2 and for $\mu := m + s$ and large n or N, we have

$$|\hat{u}_n^C| \le Cn^{-2\mu+1} U_{-1+}^{\mu,k}, \qquad \|u - \pi_N^C u\|_{L^\infty(\Omega)} \le CN^{-2\mu+2} U_{-1+}^{\mu,k}. \tag{4.20}$$

Table 1 Order of $|\hat{u}_n^C|$ with $u = (1+x)^\alpha \sin x$

n	$\alpha = 0.1$	Order	$\alpha = 1.2$	Order	$\alpha = 2.3$	Order
2^3	1.18e−02	–	3.79e−04	–	6.06e−05	–
2^4	5.10e−03	1.21	3.37e−05	3.49	1.05e−06	5.85
2^5	2.21e−03	1.21	3.14e−06	3.43	2.06e−08	5.67
2^6	9.59e−04	1.21	2.96e−07	3.41	4.18e−10	5.62
2^7	4.14e−04	1.21	2.80e−08	3.40	8.59e−12	5.60
2^8	1.77e−04	1.23	2.65e−09	3.40	1.77e−13	5.60
2^9	7.35e−05	1.27	2.51e−10	3.40	3.65e−15	5.60
Pred		1.20		3.40		5.60

Table 2 Order of $\|u - \pi_N^C u\|_{L^\infty}$ with $u = (1+x)^\alpha \sin x$

N	$\alpha = 0.1$	Order	$\alpha = 1.2$	Order	$\alpha = 2.3$	Order
2^3	4.63e−01	–	1.04e−03	–	7.27e−05	–
2^4	4.05e−01	0.19	2.06e−04	2.34	3.08e−06	4.56
2^5	3.54e−01	0.20	4.02e−05	2.36	1.32e−07	4.54
2^6	3.09e−01	0.20	7.74e−06	2.38	5.60e−09	4.56
2^7	2.69e−01	0.20	1.48e−06	2.39	2.35e−10	4.58
2^8	2.35e−01	0.19	2.81e−07	2.39	9.76e−12	4.59
2^9	2.07e−01	0.19	5.35e−08	2.40	4.04e−13	4.59
Pred		0.20		2.40		4.60

For $u(x) = (1+x)^\alpha \sin x$ with $\alpha > -1/2$, we have $\mu = \alpha + 1$, $k = [\mu]$,

$$|\hat{u}_n^C| \le C n^{-2\alpha-1} U_{-1+}^{\mu,k}, \qquad \|u - \pi_N^C u\|_{L^\infty(\Omega)} \le C N^{-2\alpha} U_{-1+}^{\mu,k},$$

where C is a positive constant independent of n, N and u.

We tabulate in Tables 1 and 2 the errors and convergence order of Chebyshev approximations for the function $u(x) = (1+x)^\alpha \sin x$ with various α, which indicate the optimal convergence order as predicted (see the last row).

Remark 4.2 For $u(x) = (1+x)^{1.2} |\ln(1+x)|^\beta$ with $\beta > -1$, we tabulate in Tables 3 and 4 the errors and convergence order of Chebyshev approximations for various β. In the first column of Tables 3 and 4, when $\beta = 0$, the equation is exactly $u(x) = (1+x)^{1.2}$, which exhibits the predicted optimal convergence order (see the last row).

We demonstrate that the estimate (4.12) is optimal for $u(x) = (1+x)^\alpha$, which is applicable to the typical singular function $u(x) = (1+x)^\alpha g(x)$ with smooth $g(x)$ and $g(-1) \ne 0$.

Theorem 4.3 *Consider $u(x) = (1+x)^\alpha$ on $\bar{\Omega} = [-1, 1]$ with $\alpha > 0$. Then we have the following bounds.*

(i) For $N > \alpha - 1$,

$$|(u - \pi_N^C u)(-1)| = \frac{|\sin(\pi\alpha)|\Gamma(2\alpha)}{2^{\alpha-1}\pi} \frac{\Gamma(N-\alpha+1)}{\Gamma(N+\alpha+1)}. \tag{4.21}$$

Table 3 Order of $|\hat{u}_n^C|$ with $u(x) = (1+x)^{1.2}|\ln(1+x)|^{\beta}$

n	$\beta = 0$	Order	$\beta = 0.2$	Order	$\beta = 1.5$	Order
2^3	4.23e−04	–	2.93e−02	–	1.06e−02	–
2^4	3.93e−05	3.43	1.29e−02	1.18	1.45e−03	2.88
2^5	3.71e−06	3.41	5.62e−03	1.20	2.22e−04	2.71
2^6	3.51e−07	3.40	2.44e−03	1.20	3.67e−05	2.60
2^7	3.32e−08	3.40	1.05e−03	1.21	6.33e−06	2.53
2^8	3.15e−09	3.40	4.50e−04	1.23	1.11e−06	2.51
2^9	2.98e−10	3.40	1.87e−04	1.27	1.97e−07	2.50
Pred		3.40		–		–

Table 4 Order of $\|u - \pi_N^C u\|_{L^\infty}$ with $u(x) = (1+x)^{1.2}|\ln(1+x)|^{\beta}$

N	$\beta = 0$	Order	$\beta = 0.2$	Order	$\beta = 1.5$	Order
2^3	1.20e−03	–	5.82e−01	–	1.39e−02	–
2^4	2.43e−04	2.30	5.13e−01	0.18	5.61e−03	1.31
2^5	4.76e−05	2.35	4.49e−01	0.19	2.10e−03	1.42
2^6	9.19e−06	2.37	3.92e−01	0.19	7.61e−04	1.46
2^7	1.76e−06	2.39	3.43e−01	0.20	2.72e−04	1.48
2^8	3.34e−07	2.39	3.00e−01	0.19	9.69e−05	1.49
2^9	6.35e−08	2.40	2.63e−01	0.19	3.45e−05	1.49
Pred		2.40		–		–

(ii) For $N > \alpha$,

$$|(u - \pi_N^C u)(1)| \le \frac{|\sin(\pi\alpha)|\Gamma(2\alpha+1)}{2^\alpha \pi} \frac{\Gamma(N-\alpha)}{\Gamma(N+\alpha+1)}. \tag{4.22}$$

(iii) For $N > \alpha + 2$,

$$|(u - \pi_N^C u)(0)| \le \frac{|\sin(\pi\alpha)|\Gamma(2\alpha+1)}{2^\alpha \pi} \frac{\Gamma(N-\alpha-2)}{\Gamma(N+\alpha-1)}. \tag{4.23}$$

Proof We first derive the explicit formula of the expansion coefficients, that is, for $n \ge \alpha+1$

$$\hat{u}_n^C = (-1)^{n+1} \widetilde{D}_\alpha \frac{\Gamma(n-\alpha)}{\Gamma(n+\alpha+1)}, \quad \widetilde{D}_\alpha = \frac{\sin(\pi\alpha)\Gamma(2\alpha+1)}{2^{\alpha-1}\pi}, \tag{4.24}$$

where we use the identity in [27, (6.16)].

(i) For $x = -1$ with $T_n(-1) = (-1)^n$, we obtain

$$(u - \pi_N^C u)(-1) = \sum_{n=N+1}^{\infty} \hat{u}_n^C T_n(-1) = -\widetilde{D}_\alpha \sum_{n=N+1}^{\infty} \frac{\Gamma(n-\alpha)}{\Gamma(n+\alpha+1)}.$$

Therefore,

$$
\begin{aligned}
|(u-\pi_N^C u)(-1)| &= \frac{|\widetilde{D}_\alpha|}{2\alpha}\sum_{n=N+1}^{\infty}\left((n+\alpha)\frac{\Gamma(n-\alpha)}{\Gamma(n+\alpha+1)}-(n-\alpha)\frac{\Gamma(n-\alpha)}{\Gamma(n+\alpha+1)}\right)\\
&= \frac{|\widetilde{D}_\alpha|}{2\alpha}\sum_{n=N+1}^{\infty}\left(\frac{\Gamma(n-\alpha)}{\Gamma(n+\alpha)}-\frac{\Gamma(n-\alpha+1)}{\Gamma(n+\alpha+1)}\right)\\
&= \frac{|\widetilde{D}_\alpha|}{2\alpha}\frac{\Gamma(N-\alpha+1)}{\Gamma(N+\alpha+1)}.
\end{aligned}
\tag{4.25}
$$

This yields (4.21).

(ii) For $x=1$ with $T_n(1)=1$, we obtain

$$
(u-\pi_N^C u)(1)=\sum_{n=N+1}^{\infty}\hat{u}_n^C T_n(1)=-\widetilde{D}_\alpha\sum_{n=N+1}^{\infty}\widetilde{\mathcal{S}}_n,\quad \widetilde{\mathcal{S}}_n=(-1)^n\frac{\Gamma(n-\alpha)}{\Gamma(n+\alpha+1)}.
$$

Thus, we have

$$
\begin{aligned}
\widetilde{\mathcal{S}}_n+\widetilde{\mathcal{S}}_{n+1} &= (-1)^n\left(\frac{\Gamma(n-\alpha)}{\Gamma(n+\alpha+1)}-\frac{\Gamma(n-\alpha+1)}{\Gamma(n+\alpha+2)}\right)\\
&= (-1)^n(2\alpha+1)\frac{\Gamma(n-\alpha)}{\Gamma(n+\alpha+2)}\\
&\le \frac{(2\alpha+1)}{2}\left(\frac{\Gamma(n-\alpha-1)}{\Gamma(n+\alpha+1)}+\frac{\Gamma(n-\alpha)}{\Gamma(n+\alpha+2)}\right)\\
&= \big(\widetilde{\mathcal{T}}_n-\widetilde{\mathcal{T}}_{n+1}\big)+\big(\widetilde{\mathcal{T}}_{n+1}-\widetilde{\mathcal{T}}_{n+2}\big),
\end{aligned}
\tag{4.26}
$$

where we denoted and noted that

$$
\begin{aligned}
&\widetilde{\mathcal{T}}_n=\frac{\Gamma(n-\alpha-1)}{2\Gamma(n+\alpha)},\quad \widetilde{\mathcal{T}}_n-\widetilde{\mathcal{T}}_{n+1}=\frac{1}{2}\left(\frac{\Gamma(n-\alpha-1)}{\Gamma(n+\alpha)}-\frac{\Gamma(n-\alpha)}{\Gamma(n+\alpha+1)}\right),\\
&\frac{\Gamma(n-\alpha)}{\Gamma(n+\alpha+2)}=\frac{(n-\alpha-1)\Gamma(n-\alpha-1)}{(n+\alpha+1)\Gamma(n+\alpha+1)}\le\frac{\Gamma(n-\alpha-1)}{\Gamma(n+\alpha+1)}.
\end{aligned}
$$

From (4.26), we obtain

$$
\begin{aligned}
|(u-\pi_N^C u)(1)| &= |\widetilde{D}_\alpha|\big\{(\widetilde{\mathcal{S}}_{N+1}+\widetilde{\mathcal{S}}_{N+2})+\cdots+(\widetilde{\mathcal{S}}_{N+2i+1}+\widetilde{\mathcal{S}}_{N+2i+2})+\cdots\big\}\\
&\le |\widetilde{D}_\alpha|\big\{(\widetilde{\mathcal{T}}_{N+1}-\widetilde{\mathcal{T}}_{N+2})+(\widetilde{\mathcal{T}}_{N+2}-\widetilde{\mathcal{T}}_{N+3})+\cdots\\
&\quad+(\widetilde{\mathcal{T}}_{N+2i+1}-\widetilde{\mathcal{T}}_{N+2i+2})+(\widetilde{\mathcal{T}}_{N+2i+2}-\widetilde{\mathcal{T}}_{N+2i+3})+\cdots\big\}\\
&= |\widetilde{D}_\alpha|\sum_{n=N+1}^{\infty}\big(\widetilde{\mathcal{T}}_n-\widetilde{\mathcal{T}}_{n+1}\big)=\frac{|\widetilde{D}_\alpha|}{2}\frac{\Gamma(N-\alpha)}{\Gamma(N+\alpha+1)}.
\end{aligned}
$$

This yields (4.22).

(iii) For $x=0$ with $T_{2k+1}(0)=0$ and $T_{2k}(0)=(-1)^k$, we have

$$
\begin{aligned}
&\hat{u}_{2k}^C=-\widetilde{D}_\alpha\frac{\Gamma(2k-\alpha)}{\Gamma(2k+\alpha+1)},\quad \widetilde{D}_\alpha=\frac{\sin(\pi\alpha)\Gamma(2\alpha+1)}{2^{\alpha-1}\pi},\\
&(u-\pi_N^C u)(0)=-\widetilde{D}_\alpha\sum_{k=\lceil\frac{N+1}{2}\rceil}^{\infty}\widetilde{\mathcal{S}}_k,\quad \widetilde{\mathcal{S}}_k=(-1)^k\frac{\Gamma(2k-\alpha)}{\Gamma(2k+\alpha+1)}.
\end{aligned}
$$

One verifies that

$$
\begin{aligned}
\widetilde{\mathcal{S}}_k + \widetilde{\mathcal{S}}_{k+1} &= (-1)^k \left(\frac{\Gamma(2k-\alpha)}{\Gamma(2k+\alpha+1)} - \frac{\Gamma(2k-\alpha+2)}{\Gamma(2k+\alpha+3)} \right) \\
&= (-1)^k \cdot 2 \cdot (2\alpha+1)(2k+1) \frac{\Gamma(2k-\alpha)}{\Gamma(2k+\alpha+3)} \\
&\le 2 \cdot (2\alpha+1) \frac{\Gamma(2k-\alpha)}{\Gamma(2k+\alpha+2)} \\
&\le \frac{2\alpha+1}{2} \left(\frac{\Gamma(2k-\alpha-3)}{\Gamma(2k+\alpha-1)} + \frac{\Gamma(2k-\alpha-2)}{\Gamma(2k+\alpha)} + \frac{\Gamma(2k-\alpha-1)}{\Gamma(2k+\alpha+1)} + \frac{\Gamma(2k-\alpha)}{\Gamma(2k+\alpha+2)} \right) \\
&= \big(\widetilde{\mathcal{T}}_{2k-3} - \widetilde{\mathcal{T}}_{2k-1}\big) + \big(\widetilde{\mathcal{T}}_{2k-1} - \widetilde{\mathcal{T}}_{2k+1}\big).
\end{aligned}
\tag{4.27}
$$

From (4.16), we have

$$
\frac{\Gamma(2k-\alpha)}{\Gamma(2k+\alpha+2)} \le \frac{\Gamma(2k-\alpha-1)}{\Gamma(2k+\alpha+1)} \le \frac{\Gamma(2k-\alpha-2)}{\Gamma(2k+\alpha)} \le \frac{\Gamma(2k-\alpha-3)}{\Gamma(2k+\alpha-1)},
$$

where we used the fact

$$
\widetilde{\mathcal{T}}_{2k} = \frac{\Gamma(2k-\alpha)}{2\Gamma(2k+\alpha+1)}, \quad \widetilde{\mathcal{T}}_{2k-3} - \widetilde{\mathcal{T}}_{2k-2} = \frac{(2\alpha+1)}{2} \frac{\Gamma(2k-\alpha-3)}{\Gamma(2k+\alpha-1)}.
$$

Thus by (4.27), we obtain

$$
\begin{aligned}
\big|(u-\pi_N^C u)(0)\big| &= |\widetilde{D}_\alpha| \Big\{ \Big(\widetilde{\mathcal{S}}_{\lceil \frac{N+1}{2} \rceil} + \widetilde{\mathcal{S}}_{\lceil \frac{N+1}{2} \rceil+1}\Big) + \cdots + \Big(\widetilde{\mathcal{S}}_{\lceil \frac{N+1}{2} \rceil+2i+1} + \widetilde{\mathcal{S}}_{\lceil \frac{N+1}{2} \rceil+2i+2}\Big) + \cdots \Big\} \\
&\le |\widetilde{D}_\alpha| \Big\{ \Big(\widetilde{\mathcal{T}}_{2\lceil \frac{N+1}{2} \rceil-3} - \widetilde{\mathcal{T}}_{2\lceil \frac{N+1}{2} \rceil-1}\Big) + \Big(\widetilde{\mathcal{T}}_{2\lceil \frac{N+1}{2} \rceil-1} - \widetilde{\mathcal{T}}_{2\lceil \frac{N+1}{2} \rceil+1}\Big) + \cdots \\
&\quad + \Big(\widetilde{\mathcal{T}}_{2\lceil \frac{N+1}{2} \rceil+2i-3} - \widetilde{\mathcal{T}}_{2\lceil \frac{N+1}{2} \rceil+2i-1}\Big) + \Big(\widetilde{\mathcal{T}}_{2\lceil \frac{N+1}{2} \rceil+2i-1} - \widetilde{\mathcal{T}}_{2\lceil \frac{N+1}{2} \rceil+2i+1}\Big) + \cdots \Big\} \\
&= |\widetilde{D}_\alpha| \sum_{k=\lceil \frac{N+1}{2} \rceil}^{\infty} \big(\widetilde{\mathcal{T}}_{2k-3} - \widetilde{\mathcal{T}}_{2k-1}\big) = \frac{|\widetilde{D}_\alpha|}{2} \frac{\Gamma(N-\alpha-2)}{\Gamma(N+\alpha-1)},
\end{aligned}
$$

where $\lceil \frac{N+1}{2} \rceil$ is the smallest integer $\ge \frac{N+1}{2}$. This yields (4.23).

□

From the pointwise error plots in Fig. 2 (left), we see the largest error occurs at the singular point $x = -1$. Indeed, the bounds in Theorem 4.3 agree well with the real decay of the errors as shown in Fig. 2 (right) and Fig. 3.

Remark 4.3 In Theorem 4.3, we present the exact error bounds of Chebyshev projections at $x = 0, \pm 1$ for $u(x) = (1+x)^\alpha$ with $\alpha > 0$. However, it is still unclear about the sharp convergence rate of Chebyshev projections at other points of the interval $[-1, 1]$. In fact, the rate of pointwise convergence of Chebyshev approximations on $[-1, 1]$ have been recently derived in [44, Thm. 2.5].

Theorem 4.4 *Consider $u(x) = (1+x)^\alpha$ on $\bar{\Omega} = [-1, 1]$ with $\alpha > 0$ and integer $N > \alpha - 1$, then we have*

$$
\|u - \pi_N^C u\|_{L^\infty(\Omega)} = C_\alpha \frac{\Gamma(N-\alpha+1)}{\Gamma(N+\alpha+1)}, \tag{4.28}
$$

where

$$
C_\alpha := \frac{|\sin(\pi\alpha)|\Gamma(2\alpha)}{2^{\alpha-1}\pi}.
$$

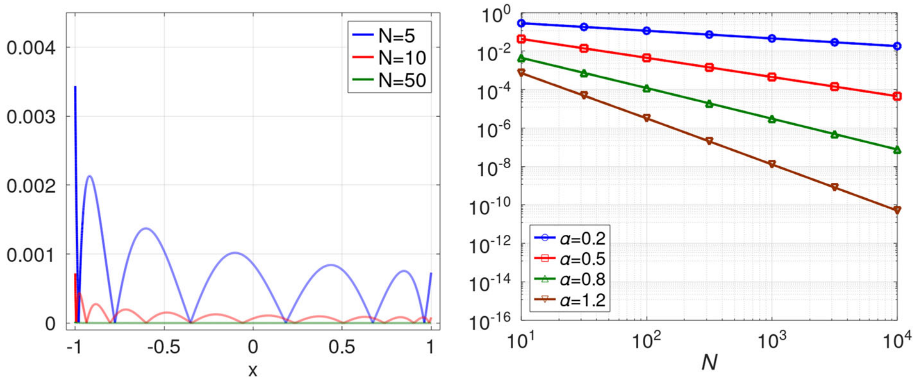

Fig. 2 Left: $|(u-\pi_N^C u)(x)|$ with $u(x)=(1+x)^{1.2}$ and different N. Right: $|(u-\pi_N^C u)(x)|$(solid line) with right bound (dotted line) in Theorem 4.3 for $x=-1$

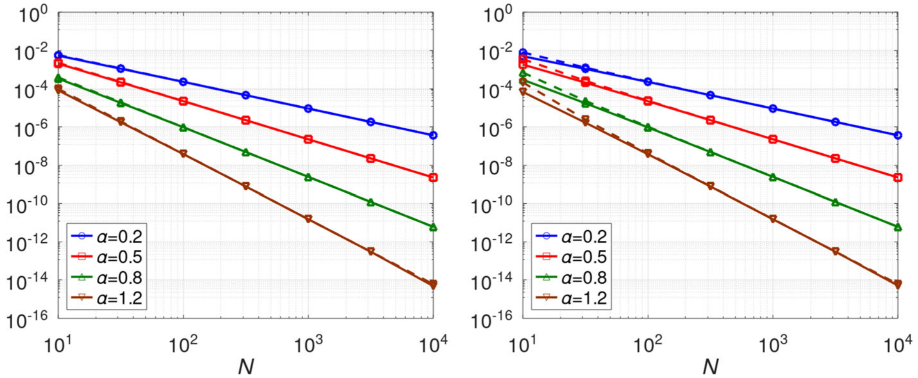

Fig. 3 $|(u-\pi_N^C u)(x)|$(solid line) with right bound (dotted line) for Theorem 4.3. Left: $x=1$; Right: $x=0$.

Proof We first derive the upper bound in (4.28), according to $|T_n(x)|\le 1$ and (4.24), we have

$$\begin{aligned}
\|u-\pi_N^C u\|_{L^\infty} &= \sum_{n=N+1}^{\infty} |\,\hat{u}_n^C\, T_n(x)| \\
&\le \sum_{n=N+1}^{\infty} |\hat{u}_n^C| = \sum_{n=N+1}^{\infty} \frac{|\sin(\pi\alpha)|\Gamma(2\alpha+1)}{2^{\alpha-1}\pi}\frac{\Gamma(n-\alpha)}{\Gamma(n+\alpha+1)} \\
&= \frac{|\sin(\pi\alpha)|\Gamma(2\alpha+1)}{2^{\alpha-1}\pi\cdot 2\alpha}\sum_{n=N+1}^{\infty}\left(\frac{\Gamma(n-\alpha)}{\Gamma(n+\alpha)}-\frac{\Gamma(n-\alpha+1)}{\Gamma(n+\alpha+1)}\right) \\
&= \frac{|\sin(\pi\alpha)|\Gamma(2\alpha)}{2^{\alpha-1}\pi}\frac{\Gamma(N-\alpha+1)}{\Gamma(N+\alpha+1)}.
\end{aligned} \tag{4.29}$$

We next derive the lower bound in (4.28) with $T_n(-1)=(-1)^n$, then we obtain from (4.25)

$$\|u-\pi_N^C u\|_{L^\infty(\Omega)} \ge |(u-\pi_N^C u)(-1)| = \frac{|\sin(\pi\alpha)|\Gamma(2\alpha)}{2^{\alpha-1}\pi}\frac{\Gamma(N-\alpha+1)}{\Gamma(N+\alpha+1)}.$$

This completes the proof. ◻

Remark 4.4 Here, we show that the Bernstein-type constant with the Chebyshev expansion

$$B_{\infty}^{(\alpha)} := \lim_{N\to\infty} N^{2\alpha} \left\| u - \pi_N^C u \right\|_{L^\infty} = \frac{|\sin(\pi\alpha)|\Gamma(2\alpha)}{2^{\alpha-1}\pi}.$$

Remark 4.5 Consequently, for $u(x) = (1+x)^\alpha$ on $\bar{\Omega} = [-1, 1]$ with $\alpha > 0$ and integer $N > \alpha - 1$, we compare the right bound of (4.12) with (4.28). Under the conditions of Theorem 4.2, for special case $u(x) = (1+x)^\alpha$, we have $\mu = \alpha + 1$, $k = [\mu]$,

$$U_{-1+}^{\mu,k} = |\sin(\pi\alpha)|\Gamma(\alpha+1),$$

thus, we obtain the right bound of (4.12)

$$\begin{aligned} M_{\infty}^{\alpha,k} = \Bigg\{ & \frac{1}{2^{\alpha+k-1}(\alpha+k)\pi} \frac{\Gamma((N-\alpha-k+1)/2)}{\Gamma((N+\alpha+k+1)/2)} \\ & + \frac{N+1}{N-\alpha-k} \frac{2^\alpha \Gamma(\alpha+1/2)}{\pi^{3/2}\alpha} \frac{\Gamma(N-\alpha+1)}{\Gamma(N+\alpha+1)} \Bigg\} |\sin(\pi\alpha)|\Gamma(\alpha+1), \end{aligned}$$

for the right bound of (4.28), we have

$$\widetilde{M}_{\infty}^{\alpha} = \frac{|\sin(\pi\alpha)|\Gamma(2\alpha)}{2^{\alpha-1}\pi} \frac{\Gamma(N-\alpha+1)}{\Gamma(N+\alpha+1)}.$$

Here, we compare right bound of (4.28) with (4.12):

$$R_{L^\infty} := \lim_{N\to\infty} \frac{\widetilde{M}_{\infty}^{\alpha}}{M_{\infty}^{\alpha,k}} = 1,$$

where we used (4.19) and $\Gamma(2z) = \pi^{-1/2} 2^{2z-1} \Gamma(z)\Gamma(z+1/2)$.

4.3 L^2-estimates of Chebyshev Expansions

As pointed out in [37, Chap. 3], the estimate of the L^2_ω-orthogonal projection is the starting point to derive many other approximation results that provide fundamental tools for error analysis of spectral and *hp* methods (see, e.g., [7, 36, 37]). Here, we present the optimal L^2_ω-estimates of the Chebyshev polynomial approximations.

Theorem 4.5 *For some $m, k \in \mathbb{N}_0$, $s \in (0, 1)$ and $\mu := m + s$, assume that $u \in \mathbb{W}^{\mu}_{-1+}(\Omega)$, $v \in \mathbb{W}^{k+1}(\Omega)$ with $v(x) = I^{1-s}_{-1+} u^{(m)}(x)$. Then we have that for $1 < \mu \le \mu + k \le N$,*

$$\begin{aligned} \|u - \pi_N^C u\|_{L^2_\omega(\Omega)} \le \Bigg\{ & \frac{4}{(2\mu+2k-1)\pi} \frac{\Gamma(N-\mu-k+1)}{\Gamma(N+\mu+k)} \\ & + \left(\frac{N+1}{N-\mu-k+1}\right)^2 \frac{2^{6\mu-4}\Gamma^2(\mu-1/2)}{\pi^2(\mu-1)(4N+3)} \frac{\Gamma(2N-2\mu+3)}{\Gamma(2N+2\mu-1)} \Bigg\}^{1/2} U_{-1+}^{\mu,k}. \end{aligned} \tag{4.30}$$

Proof Using (4.1), (4.4) and the Cauchy's inequality, we have

$$
\begin{aligned}
\|u-\pi_N^C u\|^2_{L^2_\omega(\Omega)} &= \Big\|{\sum_{n=0}^{\infty}}' \hat{u}_n^C T_n(x) - {\sum_{n=0}^{N}}' \hat{u}_n^C T_n(x)\Big\|^2_{L^2_\omega(\Omega)} = \frac{\pi}{2}\sum_{n=N+1}^{\infty}|\hat{u}_n^C|^2 \\
&\le \sum_{n=N+1}^{\infty}\left\{\frac{1}{2^{2(\mu+k-1)}\pi}\frac{\Gamma^2((n-\mu-k+1)/2)}{\Gamma^2((n+\mu+k+1)/2)}\right. \\
&\quad \left. + \frac{2^{2\mu}\Gamma^2(\mu-1/2)}{\pi^2}\frac{n^2\Gamma^2(n-\mu+1)}{(n-\mu-k)^2\Gamma^2(n+\mu)}\right\}(U^{\mu,k}_{-1+})^2.
\end{aligned}
\tag{4.31}
$$

Let us prove that

$$
\sum_{n=N+1}^{\infty}\frac{\Gamma^2((n-\mu-k+1)/2)}{\Gamma^2((n+\mu+k+1)/2)} \le \frac{2^{2(\mu+k)}}{2(\mu+k)-1}\frac{\Gamma(N-\mu-k+1)}{\Gamma(N+\mu+k)}. \tag{4.32}
$$

By (4.16), we obtain

$$
\frac{\Gamma((n-\rho+1)/2)}{\Gamma((n+\rho+1)/2)} \le \frac{\Gamma((n-\rho)/2)}{\Gamma((n+\rho)/2)}.
$$

Using the identity:

$$
\Gamma(2z) = \pi^{-1/2}2^{2z-1}\Gamma(z)\Gamma(z+1/2), \tag{4.33}
$$

we derive

$$
\begin{aligned}
\frac{\Gamma^2((n-\rho+1)/2)}{\Gamma^2((n+\rho+1)/2)} &\le \frac{\Gamma((n-\rho+1)/2)}{\Gamma((n+\rho+1)/2)}\frac{\Gamma((n-\rho)/2)}{\Gamma((n+\rho)/2)} = 2^{2\rho}\frac{\Gamma(n-\rho)}{\Gamma(n+\rho)} \\
&= \frac{2^{2\rho}}{2\rho-1}\left\{(n+\rho-1)\frac{\Gamma(n-\rho)}{\Gamma(n+\rho)} - (n-\rho)\frac{\Gamma(n-\rho)}{\Gamma(n+\rho)}\right\} \\
&= \frac{2^{2\rho}}{2\rho-1}\left\{\frac{\Gamma(n-\rho)}{\Gamma(n+\rho-1)} - \frac{\Gamma(n-\rho+1)}{\Gamma(n+\rho)}\right\}.
\end{aligned}
\tag{4.34}
$$

We can obtain (4.32) by summing up (4.34) with $\rho=\mu+k$.

Through (4.16) and (4.33), we have

$$
\begin{aligned}
\frac{n\Gamma^2(n-\mu+1)}{\Gamma^2(n+\mu)} &= \frac{n}{n+\mu-1}\frac{\Gamma^2(n-\mu+1)}{\Gamma(n+\mu-1)\Gamma(n+\mu)} \\
&\le \frac{n}{n+\mu-1}\frac{\Gamma(n-\mu+1/2)\Gamma(n-\mu+1)}{\Gamma(n+\mu-1)\Gamma(n+\mu-1/2)} \\
&= 2^{4\mu-3}\frac{n}{n+\mu-1}\frac{\Gamma(2n-2\mu+1)}{\Gamma(2n+2\mu-2)} \\
&\le \frac{2^{4\mu-6}}{\mu-1}\frac{n}{n-1/4}\left\{\frac{\Gamma(2n-2\mu+1)}{\Gamma(2n+2\mu-3)} - \frac{\Gamma(2n-2\mu+3)}{\Gamma(2n+2\mu-1)}\right\},
\end{aligned}
$$

together with (4.31) and (4.32) implies (4.30). This completes the proof. □

Remark 4.6 Using (4.19) under the conditions of Theorem 4.5 and for $\mu := m+s$ and large N, we have

$$
\|u-\pi_N^C u\|_{L^2_\omega(\Omega)} \le CN^{-2\mu+3/2}U^{\mu,k}_{-1+}. \tag{4.35}
$$

Table 5 Order of $\|u-\pi_N^C u\|_{L^2_\omega}$ with $u=(1+x)^\alpha \sin x$

N	$\alpha=0.1$	order	$\alpha=1.2$	order	$\alpha=2.3$	order
2^3	3.36e−02	–	4.49e−04	–	4.58e−05	–
2^4	2.11e−02	0.68	6.35e−05	2.82	1.38e−06	5.06
2^5	1.31e−02	0.69	8.81e−06	2.85	4.19e−08	5.04
2^6	8.04e−03	0.70	1.20e−06	2.87	1.26e−09	5.06
2^7	4.95e−03	0.70	1.63e−07	2.88	3.74e−11	5.07
2^8	3.09e−03	0.68	2.19e−08	2.89	1.10e−12	5.09
2^9	2.04e−03	0.60	2.95e−09	2.90	3.22e−14	5.09
Pred		0.70		2.90		5.10

Table 6 Order of $\|u-\pi_N^C u\|_{L^2_\omega}$ with $u(x)=(1+x)^{1.2}|\ln(1+x)|^\beta$

N	$\beta=0$	order	$\beta=0.2$	order	$\beta=1.5$	order
2^3	5.12e−04	–	5.83e−02	–	6.52e−03	–
2^4	7.45e−05	2.78	3.71e−02	0.65	1.81e−03	1.85
2^5	1.04e−05	2.84	2.33e−02	0.68	4.85e−04	1.90
2^6	1.43e−06	2.87	1.44e−02	0.69	1.26e−04	1.95
2^7	1.94e−07	2.88	8.89e−03	0.70	3.19e−05	1.98
2^8	2.61e−08	2.89	5.55e−03	0.68	8.02e−06	1.99
2^9	3.50e−09	2.90	3.67e−03	0.60	1.98e−06	2.02
Pred		2.90		–		–

Thus, for $u(x)=(1+x)^\alpha \sin x$ with $\alpha>-1/2$, we have $\mu=\alpha+1$ and $k=[\mu]$,

$$\left\|u-\pi_N^C u\right\|_{L^2_\omega(\Omega)} \leq C N^{-2\alpha-1/2} U_{-1+}^{\mu,k},$$

where C is a positive constant independent of N and u.

We tabulate in Table 5 the errors and convergence order of Chebyshev approximations in L^2-norm for the function $u(x)=(1+x)^\alpha \sin x$ with various α, which indicate the optimal convergence order as predicted (see the last row).

Remark 4.7 For $u(x)=(1+x)^{1.2}|\ln(1+x)|^\beta$ with $\beta>-1$, we tabulate in Table 6 the errors and convergence order of Chebyshev approximations in L^2-norm for various β. In the first column of Table 6, when $\beta=0$, the equation is exactly $u(x)=(1+x)^{1.2}$, which exhibits the predicted optimal convergence order (see the last row).

Theorem 4.6 *Consider* $u(x)=(1+x)^\alpha$ *on* $\bar{\Omega}=[-1,1]$ *with* $\alpha>0$ *and integer* $N>\alpha-1$, *then we have*

$$\left\|u-\pi_N^C u\right\|_{L^2_\omega(\Omega)} = \left(\tilde{C}_\alpha \frac{\Gamma(2N-2\alpha+1)}{\Gamma(2N+2\alpha+1)}\right)^{1/2}, \tag{4.36}$$

where

$$\widetilde{C}_\alpha := \frac{\alpha}{N+\alpha+1}\frac{2^{2\alpha+2}\sin^2(\pi\alpha)\Gamma^2(2\alpha)}{\pi}.$$

Proof According to (4.1) and (4.24), we have

$$
\begin{aligned}
\left\|u-\pi_N^C u\right\|_{L_\omega^2(\Omega)}^2 &= \frac{\pi}{2} \sum_{n=N+1}^{\infty} |\hat{u}_n^C|^2 \\
&= \frac{\pi}{2} \sum_{n=N+1}^{\infty} \left(\frac{|\sin(\pi\alpha)|\Gamma(2\alpha+1)}{2^{\alpha-1}\pi} \right)^2 \frac{\Gamma^2(n-\alpha)}{\Gamma^2(n+\alpha+1)}.
\end{aligned}
\tag{4.37}
$$

Through (4.16) and (4.33), we have

$$
\begin{aligned}
\frac{\Gamma^2(n-\alpha)}{\Gamma^2(n+\alpha+1)} &= \frac{1}{n+\alpha} \frac{\Gamma^2(n-\alpha)}{\Gamma(n+\alpha)\Gamma(n+\alpha+1)} \\
&\le \frac{1}{n+\alpha} \frac{\Gamma(n-\alpha-1/2)\Gamma(n-\alpha)}{\Gamma(n+\alpha)\Gamma(n+\alpha+1/2)} \\
&= \frac{2^{4\alpha+1}}{n+\alpha} \frac{\Gamma(2n-2\alpha-1)}{\Gamma(2n+2\alpha)} \\
&= \frac{2^{4\alpha-1}}{\alpha(n+\alpha)} \left\{ \frac{\Gamma(2n-2\alpha-1)}{\Gamma(2n+2\alpha-1)} - \frac{\Gamma(2n-2\alpha)}{\Gamma(2n+2\alpha)} \right\} \\
&\le \frac{2^{4\alpha-1}}{\alpha(n+\alpha)} \left\{ \frac{\Gamma(2n-2\alpha-1)}{\Gamma(2n+2\alpha-1)} - \frac{\Gamma(2n-2\alpha+1)}{\Gamma(2n+2\alpha+1)} \right\},
\end{aligned}
\tag{4.38}
$$

together with (4.37) implies (4.36). This completes the proof. □

Remark 4.8 Here, we show that the Bernstein-type constant with the Chebyshev expansion

$$
B_2^{(\alpha)} := \lim_{N\to\infty} N^{2\alpha+1/2} \left\|u-\pi_N^C u\right\|_{L_\omega^2(\Omega)} = \sqrt{\frac{\alpha}{\pi}} 2^{\alpha+1} |\sin(\pi\alpha)|\Gamma(2\alpha).
$$

Remark 4.9 Consequently, for $u(x) = (1+x)^\alpha$ on $\bar{\Omega} = [-1, 1]$ with $\alpha > 0$ and integer $N > \alpha - 1$, we compare the right bound of (4.30) with (4.36). Under the conditions of Theorem 4.5, for special case $u(x) = (1+x)^\alpha$, we have $\mu = \alpha + 1$, $k = [\mu]$, we obtain the right bound of (4.30)

$$
\begin{aligned}
M_2^{\alpha,k} = \Bigg\{ & \frac{4}{(2\alpha+2k+1)\pi} \frac{\Gamma(N-\alpha-k)}{\Gamma(N+\alpha+k+1)} \\
&+ \left(\frac{N+1}{N-\alpha-k} \right)^2 \frac{2^{6\alpha+2}\Gamma^2(\alpha+1/2)}{\pi^2\alpha(4N+3)} \frac{\Gamma(2N-2\alpha+1)}{\Gamma(2N+2\alpha+1)} \Bigg\}^{1/2} |\sin(\pi\alpha)|\Gamma(\alpha+1),
\end{aligned}
$$

for the right bound of (4.36), we have

$$
\widetilde{M}_2^\alpha = \left(\frac{\alpha}{N+\alpha+1} \frac{2^{2\alpha+2}\sin^2(\pi\alpha)\Gamma^2(2\alpha)}{\pi} \frac{\Gamma(2N-2\alpha+1)}{\Gamma(2N+2\alpha+1)} \right)^{1/2}.
$$

Here, we compare right bound of (4.36) with (4.30):

$$
R_{L_\omega^2(\Omega)} := \lim_{N\to\infty} \frac{\widetilde{M}_2^\alpha}{M_2^{\alpha,k}} = 1,
$$

where we used (4.19) and $\Gamma(2z) = \pi^{-1/2} 2^{2z-1} \Gamma(z)\Gamma(z+1/2)$.

5 Analysis of Interpolation and Quadrature

It is known that the error analysis of several widely used interpolations and quadrature boils down to estimating the coefficients $\{\hat{u}_n^C\}$ and their partial sums (cf. [29, 39, 40, 48]). Here, we consider the related interpolation and quadrature of functions at Chebyshev-Gauss points and present sharp bounds by using our new estimates on the decay of expansion coefficients.

Interpolation and quadrature at Chebyshev–Gauss (CG) points $\{x_j\}_{j=0}^N$, i.e., zeros of $T_{N+1}(x)$:

$$(\mathcal{I}_N^C u)(x) = {\sum_{n=0}^{N}}' b_n T_n(x), \quad b_n = \frac{2}{N+1}\sum_{j=0}^{N} u(x_j)T_n(x_j),$$

and

$$\int_{-1}^{1} u(x)(1-x^2)^{-1/2}\mathrm{d}x = \frac{\pi}{N+1}\sum_{j=0}^{N} u(x_j) + \mathcal{R}_N^C[u].$$

In the following theorem, we derive precise bounds for the interpolation and quadrature of functions at Chebyshev-Gauss points.

Theorem 5.1 *For some $m, k \in \mathbb{N}_0$, $s \in (0,1)$ and $\mu := m+s$, assume that $u \in \mathbb{W}^{\mu}_{-1+}(\Omega)$, $v \in \mathbb{W}^{k+1}(\Omega)$ with $v(x) = I_{-1+}^{1-s}u^{(m)}(x)$. Then for $1 < \mu \le \mu + k \le N$, we have*

$$\begin{aligned}\|u - \mathcal{I}_N^C u\|_{L^\infty(\Omega)} \le \bigg\{&(N+1)\Phi + \frac{1}{2^{\mu+k-2}(\mu+k-1)\pi}\frac{\Gamma((N-\mu-k)/2+1)}{\Gamma((N+\mu+k)/2)} \\ &+ \frac{N+1}{N-\mu-k+1}\frac{2^{\mu-1}\Gamma(\mu-1/2)}{\pi^{3/2}(\mu-1)}\frac{\Gamma(N-\mu+2)}{\Gamma(N+\mu)}\bigg\}U^{\mu,k}_{-1+},\end{aligned} \tag{5.1}$$

$$\begin{aligned}\|u - \mathcal{I}_N^C u\|_{L^2_\omega(\Omega)} \le \bigg\{&\frac{\pi(N+1)}{2}\Phi^2 + \bigg(\frac{4}{(2\mu+2k-1)\pi}\frac{\Gamma(N-\mu-k+1)}{\Gamma(N+\mu+k)} \\ &+ \Big(\frac{N+1}{N-\mu-k+1}\Big)^2\frac{2^{6\mu-4}\Gamma^2(\mu-1/2)}{\pi^2(\mu-1)(4N+3)}\frac{\Gamma(2N-2\mu+3)}{\Gamma(2N+2\mu-1)}\bigg)\bigg\}^{1/2} U^{\mu,k}_{-1+},\end{aligned} \tag{5.2}$$

and

$$\begin{aligned}|\mathcal{R}_N^C[u]| \le \bigg\{&\frac{1}{2^{\mu+k-1}}\frac{\mu+k}{\mu+k-1}\frac{\Gamma((2N-\mu-k)/2+1)}{\Gamma((2N+\mu+k)/2+1)} + \frac{2\mu-1}{\mu-1} \\ &\times \frac{N+1}{2N-\mu-k+2}\frac{2^{\mu}\Gamma(\mu-1/2)}{\sqrt{\pi}}\frac{\Gamma(2N-\mu+3)}{\Gamma(2N+\mu+2)}\bigg\}U^{\mu,k}_{-1+},\end{aligned} \tag{5.3}$$

where

$$\Phi = \Big(\frac{1}{2^{\mu+k-2}\pi}\frac{\mu+k}{\mu+k-1}\frac{\Gamma((N-\mu-k)/2+1)}{\Gamma((N+\mu+k)/2+1)} + \frac{2\mu-1}{\mu-1}\frac{N+2}{N+2-\mu-k}\frac{2^{\mu}\Gamma(\mu-1/2)}{\pi^{3/2}}\frac{\Gamma(N-\mu+3)}{\Gamma(N+\mu+2)}\Big).$$

Proof We first give a detailed proof of the L^2_ω-error of the CG interpolation. Note that

$$\begin{aligned}\mathcal{I}_N^C u(x) - u(x) &= \mathcal{I}_N^C u(x) - \pi_N^C u(x) + \pi_N^C u(x) - u(x) \\ &= {\sum_{n=0}^{N}}'(b_n - \hat{u}_n^C)T_n(x) + \pi_N^C u(x) - u(x).\end{aligned} \tag{5.4}$$

Hence, we obtain

$$\|u - \mathcal{I}_N^C u\|^2_{L^2_\omega(\Omega)} \le \frac{\pi}{2}{\sum_{n=0}^{N}}'|b_n - \hat{u}_n^C|^2 + \|u - \pi_N^C u\|^2_{L^2_\omega(\Omega)}. \tag{5.5}$$

Recall that (cf. [8, (4.56)]):

$$b_n-\hat{u}_n^C=\sum_{k=1}^{\infty}(-1)^k\big(\hat{u}^C_{2k(N+1)-n}+\hat{u}^C_{2k(N+1)+n}\big),\quad n=0,\cdots,N,$$

we denote $\widetilde{N}=2(N+1)$ and derive from (4.4)

$$\big|b_n-\hat{u}_n^C\big|\le\sum_{i=1}^{\infty}\big\{|\hat{u}^C_{i\widetilde{N}-n}|+|\hat{u}^C_{i\widetilde{N}+n}|\big\}\le\big\{Q_1+Q_2\big\}U^{\mu,k}_{-1+},\tag{5.6}$$

where

$$Q_1=\frac{1}{2^{\mu+k-1}\pi}\sum_{i=1}^{\infty}\left\{\frac{\Gamma((i\widetilde{N}-n-\mu-k+1)/2)}{\Gamma((i\widetilde{N}-n+\mu+k+1)/2)}+\frac{\Gamma((i\widetilde{N}+n-\mu-k+1)/2)}{\Gamma((i\widetilde{N}+n+\mu+k+1)/2)}\right\},$$
$$Q_2=\frac{2^{\mu}\Gamma(\mu-1/2)}{\pi^{3/2}}\sum_{i=1}^{\infty}\left\{\frac{i\widetilde{N}-n}{i\widetilde{N}-n-\mu-k}\frac{\Gamma(i\widetilde{N}-n-\mu+1)}{\Gamma(i\widetilde{N}-n+\mu)}+\frac{i\widetilde{N}+n}{i\widetilde{N}+n-\mu-k}\frac{\Gamma(i\widetilde{N}+n-\mu+1)}{\Gamma(i\widetilde{N}+n+\mu)}\right\}.$$

Taking $a=0$, and $b=\mu+k$ in (4.16), we have

$$\frac{\Gamma((i\widetilde{N}+n-\mu-k+1)/2)}{\Gamma((i\widetilde{N}+n+\mu+k+1)/2)}\le\frac{\Gamma((i\widetilde{N}-n-\mu-k+1)/2)}{\Gamma((i\widetilde{N}-n+\mu+k+1)/2)},$$

then

$$Q_1\le\frac{1}{2^{\mu+k-2}\pi}\sum_{i=1}^{\infty}\frac{\Gamma((i\widetilde{N}-n-\mu-k+1)/2)}{\Gamma((i\widetilde{N}-n+\mu+k+1)/2)}.\tag{5.7}$$

Now, we consider how to estimate Q_1, for convenience, we denote

$$\mathbb{S}_m^{\alpha}:=\frac{\Gamma((\widetilde{N}+m-n-\alpha+1)/2)}{\Gamma((\widetilde{N}+m-n+\alpha+1)/2)},\quad\mathbb{T}_m^{\alpha}:=\frac{\Gamma((\widetilde{N}+m-n-\alpha+1)/2)}{\Gamma((\widetilde{N}+m-n+\alpha-1)/2)},\quad\alpha=\mu+k.$$

From direct calculations similar to (4.14), we obtain $\mathbb{T}_m^{\alpha}-\mathbb{T}_{m+2}^{\alpha}=(\alpha-1)\mathbb{S}_m^{\alpha}$. Thus, we readily conclude that

$$\begin{aligned}\sum_{i=1}^{\infty}\frac{\Gamma((i\widetilde{N}-n-\alpha+1)/2)}{\Gamma((i\widetilde{N}-n+\alpha+1)/2)}&=\frac{\Gamma((\widetilde{N}-n-\alpha+1)/2)}{\Gamma((\widetilde{N}-n+\alpha+1)/2)}+\sum_{i=2}^{\infty}\frac{\Gamma((i\widetilde{N}-n-\alpha+1)/2)}{\Gamma((i\widetilde{N}-n+\alpha+1)/2)}\\&\le\frac{\Gamma((\widetilde{N}-n-\alpha+1)/2)}{\Gamma((\widetilde{N}-n+\alpha+1)/2)}+\frac{1}{\widetilde{N}}\sum_{i=2}^{\infty}\sum_{m=(i-2)\widetilde{N}+1}^{(i-1)\widetilde{N}}\mathbb{S}_m^{\alpha}\\&=\frac{\Gamma((\widetilde{N}-n-\alpha+1)/2)}{\Gamma((\widetilde{N}-n+\alpha+1)/2)}+\frac{1}{\widetilde{N}}\frac{1}{(\alpha-1)}\sum_{i=2}^{\infty}\big(\mathbb{T}_1^{\alpha}+\mathbb{T}_2^{\alpha}-\mathbb{T}_3^{\alpha}\\&\qquad-\mathbb{T}_4^{\alpha}+\mathbb{T}_3^{\alpha}-\cdots+\mathbb{T}^{\alpha}_{(i-2)\widetilde{N}+1}-\mathbb{T}^{\alpha}_{(i-2)\widetilde{N}+3}+\cdots\big)\\&\le\frac{\Gamma((\widetilde{N}-n-\alpha+1)/2)}{\Gamma((\widetilde{N}-n+\alpha+1)/2)}+\frac{1}{\widetilde{N}}\frac{1}{(\alpha-1)}\big(\mathbb{T}_1^{\alpha}+\mathbb{T}_2^{\alpha}\big).\end{aligned}\tag{5.8}$$

Using (4.16), we have $\mathbb{T}_2^{\alpha}\le\mathbb{T}_1^{\alpha}$, together with (5.7) and (5.8), we obtain

$$\begin{aligned}Q_1&\le\frac{1}{2^{\mu+k-2}\pi}\left(\frac{\Gamma((\widetilde{N}-n-\mu-k+1)/2)}{\Gamma((\widetilde{N}-n+\mu+k+1)/2)}+\frac{\widetilde{N}-n-\mu-k}{\widetilde{N}(\mu+k-1)}\frac{\Gamma((\widetilde{N}-n-\mu-k)/2)}{\Gamma((\widetilde{N}-n+\mu+k)/2)}\right)\\&\le\frac{1}{2^{\mu+k-2}\pi}\frac{\mu+k}{\mu+k-1}\frac{\Gamma((\widetilde{N}-n-\mu-k)/2)}{\Gamma((\widetilde{N}-n+\mu+k)/2)}.\end{aligned}\tag{5.9}$$

Next, we consider about how to estimate Q_2, again for notational convenience, we denote

$$\widetilde{\mathbb{S}}_m := \frac{\Gamma(\widetilde{N}+m-n-\mu+1)}{\Gamma(\widetilde{N}+m-n+\mu)}, \quad \widetilde{\mathbb{T}}_m := \frac{\Gamma(\widetilde{N}+m-n-\mu+1)}{\Gamma(\widetilde{N}+m-n+\mu-1)},$$

analogous to (4.14), we obtain $\widetilde{\mathbb{T}}_m - \widetilde{\mathbb{T}}_{m+1} = 2(\mu-1)\widetilde{\mathbb{S}}_m$. Therefore, we can get

$$\begin{aligned}\sum_{i=1}^{\infty} \frac{\Gamma(i\widetilde{N}-n-\mu+1)}{\Gamma(i\widetilde{N}-n+\mu)} &= \frac{\Gamma(\widetilde{N}-n-\mu+1)}{\Gamma(\widetilde{N}-n+\mu)} + \sum_{i=2}^{\infty} \frac{\Gamma(i\widetilde{N}-n-\mu+1)}{\Gamma(i\widetilde{N}-n+\mu)} \\ &\le \frac{\Gamma(\widetilde{N}-n-\mu+1)}{\Gamma(\widetilde{N}-n+\mu)} + \frac{1}{\widetilde{N}} \sum_{i=2}^{\infty} \sum_{m=(i-2)\widetilde{N}+1}^{(i-1)\widetilde{N}} \widetilde{\mathbb{S}}_m \\ &= \frac{\Gamma(\widetilde{N}-n-\mu+1)}{\Gamma(\widetilde{N}-n+\mu)} + \frac{1}{\widetilde{N}} \frac{1}{2(\mu-1)} \sum_{i=2}^{\infty} (\widetilde{\mathbb{T}}_1 - \widetilde{\mathbb{T}}_2 + \widetilde{\mathbb{T}}_2 \\ &\quad - \widetilde{\mathbb{T}}_3 + \cdots + \widetilde{\mathbb{T}}_{(i-2)\widetilde{N}+1} - \widetilde{\mathbb{T}}_{(i-2)\widetilde{N}+2} + \cdots) \\ &= \frac{\Gamma(\widetilde{N}-n-\mu+1)}{\Gamma(\widetilde{N}-n+\mu)} + \frac{1}{\widetilde{N}} \frac{1}{2(\mu-1)} \widetilde{\mathbb{T}}_1 \\ &\le \frac{2\mu-1}{2(\mu-1)} \frac{\Gamma(\widetilde{N}-n-\mu+1)}{\Gamma(\widetilde{N}-n+\mu)}. \end{aligned} \tag{5.10}$$

Similarly,

$$\sum_{i=1}^{\infty} \frac{\Gamma(i\widetilde{N}+n-\mu+1)}{\Gamma(i\widetilde{N}+n+\mu)} \le \frac{2\mu-1}{2(\mu-1)} \frac{\Gamma(\widetilde{N}+n-\mu+1)}{\Gamma(\widetilde{N}+n+\mu)}. \tag{5.11}$$

From (5.6), (5.9), (5.10) and (5.11), we obtain

$$\begin{aligned}|b_n - \hat{u}_n^C| \le &\left\{ \frac{1}{2^{\mu+k-2}\pi} \frac{\mu+k}{\mu+k-1} \frac{\Gamma((\widetilde{N}-n-\mu-k)/2)}{\Gamma((\widetilde{N}-n+\mu+k)/2)} + \frac{2\mu-1}{2(\mu-1)} \frac{2^{\mu}\Gamma(\mu-1/2)}{\pi^{3/2}} \right. \\ &\left. \left(\frac{\widetilde{N}-n}{\widetilde{N}-n-\mu-k} \frac{\Gamma(\widetilde{N}-n-\mu+1)}{\Gamma(\widetilde{N}-n+\mu)} + \frac{\widetilde{N}+n}{\widetilde{N}+n-\mu-k} \frac{\Gamma(\widetilde{N}+n-\mu+1)}{\Gamma(\widetilde{N}+n+\mu)} \right) \right\} U_{-1+}^{\mu,k} \\ \le &\left\{ \frac{1}{2^{\mu+k-2}\pi} \frac{\mu+k}{\mu+k-1} \frac{\Gamma((\widetilde{N}-n-\mu-k)/2)}{\Gamma((\widetilde{N}-n+\mu+k)/2)} + \frac{2\mu-1}{\mu-1} \frac{2^{\mu}\Gamma(\mu-1/2)}{\pi^{3/2}} \right. \\ &\left. \times \frac{\widetilde{N}-n}{\widetilde{N}-n-\mu-k} \frac{\Gamma(\widetilde{N}-n-\mu+1)}{\Gamma(\widetilde{N}-n+\mu)} \right\} U_{-1+}^{\mu,k}, \end{aligned} \tag{5.12}$$

where we used (4.16).

Finally, substituting (4.30) and (5.12) in (5.5), we obtain

$$\begin{aligned}\|u - \mathcal{I}_N^C u\|_{L^2_\omega(\Omega)} \le &\left\{ \frac{\pi(N+1)}{2} \left(\frac{1}{2^{\mu+k-2}\pi} \frac{\mu+k}{\mu+k-1} \frac{\Gamma((N-\mu-k)/2+1)}{\Gamma((N+\mu+k)/2+1)} + \frac{2\mu-1}{\mu-1} \right. \right. \\ &\left. \times \frac{N+2}{N+2-\mu-k} \frac{2^{\mu}\Gamma(\mu-1/2)}{\pi^{3/2}} \frac{\Gamma(N-\mu+3)}{\Gamma(N+\mu+2)} \right)^2 \\ &+ \left(\frac{4}{(2\mu+2k-1)\pi} \frac{\Gamma(N-\mu-k+1)}{\Gamma(N+\mu+k)} \right. \\ &\left. \left. + \left(\frac{N+1}{N-\mu-k+1} \right)^2 \frac{2^{6\mu-4}\Gamma^2(\mu-1/2)}{\pi^2(\mu-1)(4N+3)} \frac{\Gamma(2N-2\mu+3)}{\Gamma(2N+2\mu-1)} \right) \right\}^{1/2} U_{-1+}^{\mu,k}. \end{aligned} \tag{5.13}$$

From (5.4), we obtain

$$\|u - \mathcal{I}_N^C u\|_{L^\infty(\Omega)} \le {\sum_{n=0}^{N}}' |b_n - \hat{u}_n^C| + \|u - \pi_N^C u\|_{L^\infty(\Omega)},$$

then by (4.12) and (5.12), we get

$$\begin{aligned}\|u-\mathcal{I}_N^C u\|_{L^\infty(\Omega)} \le \Bigg\{(N+1)\Bigg\{&\frac{1}{2^{\mu+k-2}\pi}\frac{\mu+k}{\mu+k-1}\frac{\Gamma((N-\mu-k)/2+1)}{\Gamma((N+\mu+k)/2+1)}+\frac{2\mu-1}{\mu-1}\\ &\times\frac{N+2}{N+2-\mu-k}\frac{2^\mu\Gamma(\mu-1/2)}{\pi^{3/2}}\frac{\Gamma(N-\mu+3)}{\Gamma(N+\mu+2)}\Bigg\}\\ &+\Bigg\{\frac{1}{2^{\mu+k-2}(\mu+k-1)\pi}\frac{\Gamma((N-\mu-k)/2+1)}{\Gamma((N+\mu+k)/2)}\\ &+\frac{N+1}{N-\mu-k+1}\frac{2^{\mu-1}\Gamma(\mu-1/2)}{\pi^{3/2}(\mu-1)}\frac{\Gamma(N-\mu+2)}{\Gamma(N+\mu)}\Bigg\}\Bigg\}U^{\mu,k}_{-1+}.\end{aligned} \tag{5.14}$$

Recall that (cf. [34, (6)]):

$$\mathcal{R}_N^C[u]=\pi\sum_{k=1}^{\infty}(-1)^k\hat{u}^C_{2k(N+1)}. \tag{5.15}$$

Thus, we have

$$|\mathcal{R}_N^C[u]|\le\pi\sum_{i=1}^{\infty}|\hat{u}^C_{2i(N+1)}|,$$

using a similar proof of (5.12), we can obtain (5.3). This completes the proof. □

Remark 5.1 Using (4.19) under the conditions of Theorem 5.1 and for $\mu := m + s$ and large N, we have the following estimates which are similar to (4.20) and (4.35)

$$\begin{aligned}&|\mathcal{R}_N^C[u]|\le CN^{-2\mu+1}U^{\mu,k}_{-1+},\\ &\|u-\mathcal{I}_N^C u\|_{L^\infty}\le CN^{-2\mu+2}U^{\mu,k}_{-1+},\quad \|u-\mathcal{I}_N^C u\|_{L^2_\omega}\le CN^{-2\mu+3/2}U^{\mu,k}_{-1+}.\end{aligned}$$

Thus, for $u(x) = (1 + x)^\alpha \sin x$ with $\alpha > -1/2$, we have $\mu = \alpha + 1$ and $k = [\mu]$,

$$\begin{aligned}&|\mathcal{R}_N^C[u]|\le CN^{-2\alpha-1}U^{\mu,k}_{-1+},\\ &\|u-\mathcal{I}_N^C u\|_{L^\infty}\le CN^{-2\alpha}U^{\mu,k}_{-1+},\quad \|u-\mathcal{I}_N^C u\|_{L^2_\omega}\le CN^{-2\alpha-1/2}U^{\mu,k}_{-1+},\end{aligned}$$

where C is a positive constant independent of N and u.

We tabulate in Tables 7, 8 and 9 the errors and convergence order of the related interpolation and quadrature for the function $u(x) = (1 + x)^\alpha \sin x$ with various α at Chebyshev-Gauss points, which indicate the optimal convergence order as predicted (see the last row).

6 Concluding Remarks

Motivated by the work of Liu et al. [27, 28], we derive the optimal decay rates of the coefficients for functions with endpoint singularities which are expanded in the form of Chebyshev polynomial series. For functions exhibiting singularities at the endpoints, we provide a detailed proof of the optimal error estimates for Chebyshev orthogonal projection in L^∞-norm and L^2-norm. Furthermore, the point-wise error estimates and the precise upper and lower bounds for $u(x) = (1 + x)^\alpha$, $\alpha > 0$ on $\bar{\Omega} = [-1, 1]$ in L^∞-norm are given. We also present a comprehensive set of optimal Chebyshev approximation results for related

Table 7 Order of $|\mathcal{R}_N^C[u]|$ with $u=(1+x)^\alpha \sin x$

N	$\alpha=0.1$	Order	$\alpha=1.2$	Order	$\alpha=2.3$	Order
2^3	1.16e−02	–	9.78e−05	–	3.24e−06	–
2^4	5.04e−03	1.20	9.10e−06	3.43	6.34e−08	5.68
2^5	2.19e−03	1.20	8.58e−07	3.41	1.29e−09	5.62
2^6	9.54e−04	1.20	8.11e−08	3.40	2.65e−11	5.61
2^7	4.15e−04	1.20	7.68e−09	3.40	5.45e−13	5.60
2^8	1.80e−04	1.20	7.28e−10	3.40	1.12e−14	5.60
2^9	7.78e−05	1.21	6.90e−11	3.40	2.32e−16	5.60
Pred		1.20		3.40		5.60

Table 8 Order of $\|u-\mathcal{I}_N^C u\|_{L^\infty}$ with $u=(1+x)^\alpha \sin x$

N	$\alpha=0.1$	Order	$\alpha=1.2$	Order	$\alpha=2.3$	Order
2^3	5.25e−01	–	1.73e−03	–	1.14e−04	–
2^4	4.62e−01	0.19	3.63e−04	2.25	5.40e−06	4.40
2^5	4.04e−01	0.19	7.31e−05	2.31	2.46e−07	4.46
2^6	3.53e−01	0.20	1.43e−05	2.35	1.08e−08	4.51
2^7	3.08e−01	0.20	2.76e−06	2.37	4.58e−10	4.55
2^8	2.68e−01	0.20	5.28e−07	2.39	1.92e−11	4.58
2^9	2.34e−01	0.20	1.01e−07	2.39	8.00e−13	4.59
Pred		0.20		2.40		4.60

Table 9 Order of $\|u-\mathcal{I}_N^C u\|_{L^2_\omega}$ with $u=(1+x)^\alpha \sin x$

N	$\alpha=0.1$	Order	$\alpha=1.2$	Order	$\alpha=2.3$	Order
2^3	4.32e−02	–	5.61e−04	–	5.21e−05	–
2^4	2.75e−02	0.65	8.43e−05	2.73	1.71e−06	4.93
2^5	1.72e−02	0.68	1.21e−05	2.80	5.53e−08	4.95
2^6	1.07e−02	0.69	1.69e−06	2.84	1.72e−09	5.01
2^7	6.56e−03	0.70	2.31e−07	2.87	5.19e−11	5.05
2^8	4.03e−03	0.70	3.13e−08	2.88	1.54e−12	5.07
2^9	2.53e−03	0.67	4.22e−09	2.89	4.54e−14	5.09
Pred		0.70		2.90		5.10

interpolation and quadrature of functions using Chebyshev–Gauss points. All these results were illustrated by numerical experiments. It is worth noting that our discussion primarily focuses on fractional power singularities, while the exploration of modifying fractional spaces to effectively characterize functions with logarithmic singularities remains open (see [27]). At the same time, hopefully the analysis techniques we used can pave the way for the numerical analysis of p-version and hp-version for singular problems.

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Data Availability The code used in this work will be made available upon request to the authors

Declarations

Conflict of interest The authors declare that they have no conflict of interest.

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